

GNSS Continuously Operating Reference Stations Development and Verification using Commercial Off the Shelf Receivers



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using Commercial Off the Shelf Receivers

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Dedication

In the Name of Allah, who knows all that is contained in the Heavens and all that is upon the Earth; and surely He has complete knowledge of everything.

I would like to dedicate this thesis to my parents, who have always been my biggest supporters and encouraged me to pursue my dreams.

I am also incredibly grateful to my supervisor, for his unwavering guidance and support throughout the project. His expertise and encouragement have been invaluable to me.

&

This thesis is dedicated to you!

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Abstract

Positioning, Navigation, and Timing (PNT) services have been revolutionized by the Global Navigation Satellite System (GNSS), which provides accurate & precise data for wide range of applications in daily life. As a result, improving accuracy and precision remains a challenge, and various techniques have developed to minimize the errors, including use of multiple constellations, multiple techniques of observation and multiple frequencies, as well as solutions of network.

Many countries around the world have adopted GNSS Continuously Operating Reference System (CORS) to manage industrial and engineering applications. CORS networks use real-time kinematics (RTK) techniques, service availability and high quality positioning depend on number of CORS stations and the distance between their interstation.

Geodetic field research is urgently needed in Pakistan, and the country currently only has one operational CORS network, which is not sufficient for coverage across the country. To increase the precision and accuracy of positioning and navigation services in Pakistan, this project aims to establish a carrier-based Continuously Operating Reference Station network. We intend to assess the effectiveness of the new network by comparing results of carrier-based RTK with those of standalone positioning. The study will provide important information for establishing and managing a comprehensive CORS network in Pakistan by expanding the scope of the research.

Its outcome will be crucial for establishment and management of carrier-based network of Continuously Operating Reference System in Pakistan, as it aims to provide better positioning and navigation services.

Keywords – Global Navigation Satellite System (GNSS), Continuously Operating Reference System (CORS), Real Time Kinematics (RTK), Positioning, Navigation, and Timing (PNT).

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1. INTRODUCTION

The history of GNSS dates back to the 1960s when the United States Navy developed the Transit system, which was the first satellite base navigation system. Transit system used low earth orbit satellites to provide positioning and navigation services to the US Navy. In the 1970s, the United States Air Force began developing the Global Positioning System (GPS) to provide positioning services globally and navigation services to military & civilian users. The GPS system became fully operational in 1994 and was available for use of civilians.

In time of 1990s, the Russian Federation have launched Global Orbiting Navigation Satellite System (GLONASS) to compete with GPS. The GLONASS system was initially designed for the use of military, but later it was made available for civilian use as well.

In the 2000s, the European Union began developing the Galileo satellite navigation system, which aimed to give better precise positioning & navigation service to users around worldwide. The Galileo system became operational in 2016.

In recent years, other countries have also developed their own satellite navigation systems, such as China's BeiDou Navigation Satellite System (BDS) and India's Indian Regional Navigation Satellite System (IRNSS).

Today, GNSS has become an essential part of various life applications, including transportation, communication, aviation, agriculture, and surveying. The evolution of GNSS has led to the development of advanced techniques and systems that have improved precision, accuracy and reliability of navigation & positioning services.

GNSS usage has rapidly increased in recent decades, and the need for more accurate measurements has grown. The development of Continuously Operating Reference Stations (CORS) for geodetic and monitoring purposes has resulted from the proposed and studied techniques by researchers, agencies, and commercial firms. To provide real-time corrections for construction engineering surveying and machine guiding, real-time kinematic (RTK) was invented in the 1990s. By providing accurate and precise positioning in real time, RTK proved revolutionary.

As a result of the distance-dependent sources of error, such as ephemeris, ionospheric, and tropospheric delays, RTK relied on receiving corrections from a nearby reference station which posed a challenge for users. As a result, the distance between stations was limited to 10-20 km in order to minimize the effects of atmospheric conditions. Over time, however, RTK became more usable over greater distances as a result of subsequent studies. It was proposed that network solutions be implemented to overcome the limitations of RTK, where the roving receiver is able to perform corrections based on a network of reference stations. By using this approach, a '24/7' network with strategic coverage is provided, delivering more reliable and efficient correction models. Additionally, for accurate and reliable RTK jobs, it is critical to have a dense network of permanent GNSS stations. There is however an additional instrument and effort required if there is no CORS. It has been proposed by influential researchers that the distance restriction can be extended to 50 kilometers and beyond.

1.1. Problem Statement

While GNSS is a widely used satellite navigation system, its standalone positioning accuracy may not meet the requirements of certain applications, such as surveying and mapping. Therefore, there is a need to improve its performance through various augmentation techniques. These techniques include using ground-based reference stations to provide correction solutions and improve the accuracy of GNSS signals. However, to ensure the optimal performance of these techniques, an effective augmentation system is required that can provide correction solutions to all user devices within its vicinity. This system should be able to analyze and correct errors in the received signals to provide precise and reliable positioning information. Achieving such an optimal GNSS augmentation system can significantly enhance the capabilities of GNSS and expand its applicability to various fields, including aviation, transportation, agriculture, and disaster management.

1.2. Proposed Solution

As a proposed solution to the problem statement mentioned above, the development of a GNSS augmentation system can be explored. The system should be designed to provide high-accuracy correction solutions to all GNSS enabled user devices within its coverage area (limited baseline). This system should be able to augment the performance of standalone GNSS by reducing errors and providing more accurate positioning solution.

One approach could be development of a Real-Time Kinematic (RTK) GNSS augmentation system, which is capable of providing centimeter-level positioning accuracy. The system could use a network of base stations that continuously transmit correction

information to user devices in the coverage area. The user devices could then use this information to improve their positioning accuracy.

1.3. Research Contribution

Once the planning stage is complete, the implementation phase of the project begins and the instructions provided in the study are used as a reference. While the scientific benefits of the project have been mentioned earlier, for example analyzing & planning expected quality of positioning, main purpose is to utilize the results in practical applications. Main goal is to have a well-covered network of CORS in the region as the first step, followed by the development of a network RTK (N-RTK or NRTK) solution system for dependent sectors in the future.

This system of positioning has applications across a broad spectrum in fields such as aviation, engineering, agriculture, transportation, and machine guidance, navigation and control. RTK and N-RTK systems are particularly in high demand in aviation, oil industry, general engineering, and geomatics engineering. The collection of spatial information is also becoming more important for GIS, cadastral mapping, and cartographic mapping. The use of GNSS RTK is beneficial both directly through use of a roving receiver, and indirectly through the use of manned and unmanned aero vehicles for aerial surveying, remote sensing, and photogrammetry. Therefore, the implementation of an efficient and accurate GNSS augmentation system can greatly benefit a wide range of industries and applications.

1.4. Thesis Structure

The thesis starts with the introduction (Chapter 1) which describes the background of GNSS, RTK, CORS, the purpose of the thesis, problem statement, the study contribution & structure of thesis. (Chapter 2) provides a literature review about the evolution of GNSS, guidelines, and relative conducted researches. (Chapter 3) Methodology, model of creating Mini CORS network using principles and instructions & statistical measures. (Chapter 4) (results and analysis) analysis of result obtained from the study. (Chapter 5) Conclusion and future recommendations.

2. LITERATURE REVIEW

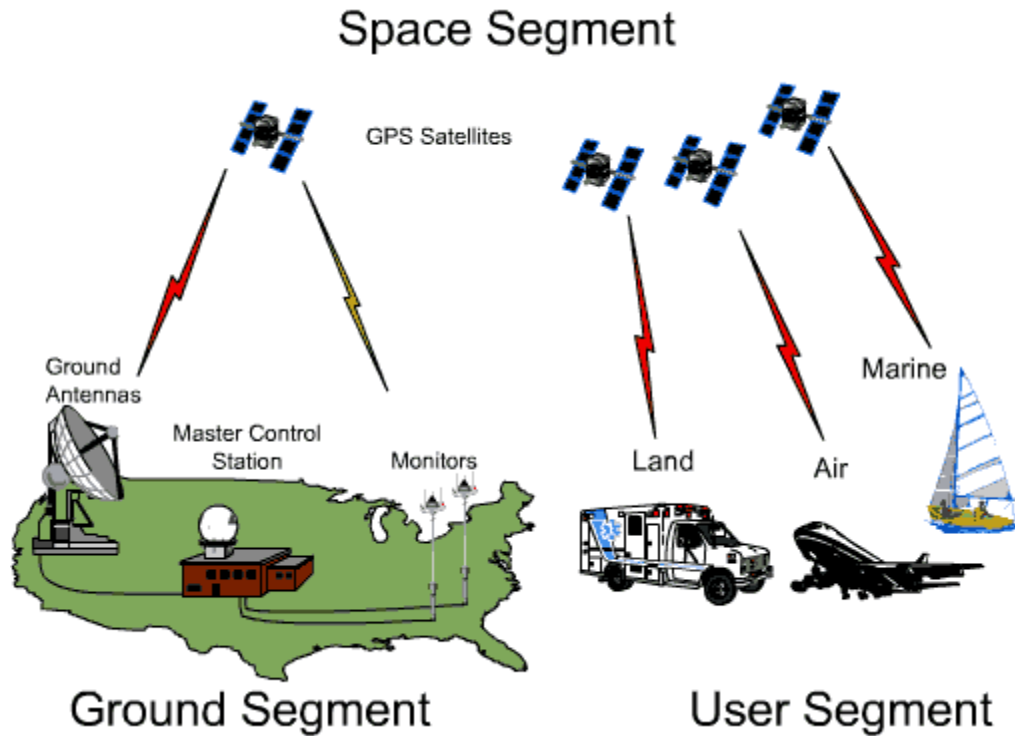
2.1. Global Navigation Satellite System

As satellite navigation systems have evolved over the years, they have undergone a significant change in technology as well. GNSS stands for Global Navigation Satellite System. It refers to satellite-based navigation systems which provide global coverage and offer navigation solutions to people around the world. However, the older systems like Transit and Tsikada, which operated in Low Earth Orbit (LEO), became obsolete due to their limited coverage and capacity to provide navigation solutions for only one visible satellite at a time. In the early 1990s, the Global Positioning System (GPS) and Global'naya Navigatsionnaya Sputnikova Sistema (GLONASS) were launched, operating in the Medium Earth Orbit (MEO) and providing more than three visible satellites at a given time, thus rendering Transit and Tsikada irrelevant.

The advancements in satellite navigation systems enabled real-time delivery of services, a feature that was previously missing in Transit and Tsikada. GNSS consists of three segments: space, ground, and the user's segments. The space segment consists of a satellite constellation arranged in a specific orbit with an inclination of orbit and located more than 20,000 km above the Earth. The ground segment, also known as the control station, comprises master control stations, data uploading stations, Antennas for GNSS satellite communication are also installed on local monitoring stations.

The user segment comprises various device & gadgets that calculate the velocity, position & time. These range from mobile phones to handheld GNSS receivers used by hikers, to specialized receivers used for high-end mapping and surveying

applications. GNSS has become an integral part of daily life, with its usage spanning across various sectors like aviation, transportation, defense, and personal navigation, among others.



At present, the four primary GNSS missions, namely GPS, GLONASS, Galileo,

Fig. 2.1 Three segments of GNSS

and BeiDou, are providing global coverage and a high degree of accuracy to users worldwide. These systems employ advanced technology to transmit signals from GNSS satellites, which are received by ground receiver stations, processed, and converted into position, velocity, and time data. The widespread availability of GNSS has revolutionized navigation and positioning across numerous sectors, including transportation, agriculture, surveying, and telecommunications.

In addition to GNSS, there are also RNSS missions, such as QZSS from Japan

and IRNSS/NavIC from India, which operate on similar principles. These systems provide regional coverage and are designed to augment the capabilities of GNSS in their respective regions. QZSS is particularly useful for navigation in urban areas with high-rise buildings, while NavIC offers excellent performance over the Indian subcontinent and its surrounding regions.

Overall, the availability of multiple GNSS and RNSS missions provides users with a range of options for navigation and positioning, ensuring global coverage and improved accuracy in a variety of applications.

Table 2.1: A brief overview of GNSS and RNSS

System	No. of Satellites	Orbits	Frequency [MHz]	Initial Service	Country	Coverage
GPS	24	MEO	L1, 1575.42 L2, 1227.60 L5, 1176.45	Dec 1993	USA	Global
GLONASS	24	MEO	L1, 1602.00 L2, 1246.00 L3, 1202.02	Sep 1993	Russia	Global
Galileo	30	MEO	E1, 1575.42 E5a, 1176.45 E5b, 1207.14 E6, 1278.75	2016-17	Europe	Global
BeiDou	35	MEO, GEO, IGSO	B1, 1561.09 B2, 1207.14 B3, 1268.52	Dec 2012	China	Global
QZSS	4	GEO, IGSO	L1, 1575.42 L2, 1227.60 L5, 1176.45 E6, 1278.75	2018	Japan	East Asia Oceania region
NavIC	7	GEO, IGSO	L5, 1176.45 S, 2492.028	2016	India	Regional

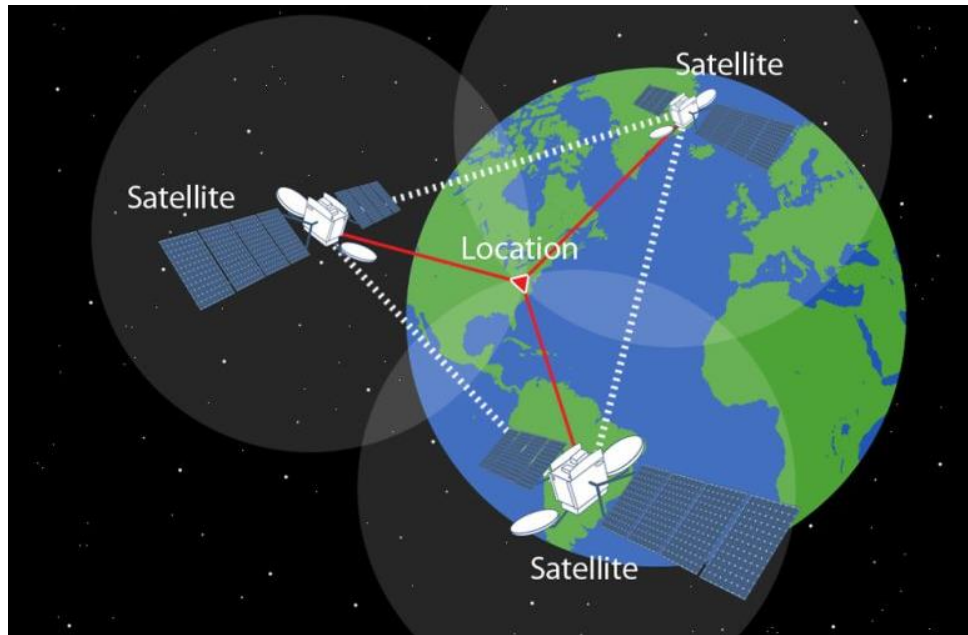


Fig. 2.2: Illustration of trilateration process of GNSS [Courtesy: UNAVCO]

The principle of trilateration is the foundation of all navigation satellite systems. Trilateration is a mathematical process used by navigation satellite to determine the accurate position of users. The process involves the determination of user position based on distances between the user and three or more satellites in space. A fourth satellite can be used for time calculation to ensure higher accuracy. The trilateration process is depicted in Figure 2.2.

To obtain the user's position accurately, each satellite transmits a signal to the user receiver. The receiver processes the signals and determines the distance between itself and each satellite based on the time when signal takes to travel from the satellite towards receiver. By determining user's distance from at least three different satellites, the user position can be calculated using mathematical algorithms based on the principles of trilateration.

This trilateration process is used by all GNSS and RNSS systems to provide accurate positioning, velocity, and time information to their users. By constantly transmitting signals from multiple satellites in space and processing these signals on the ground, these systems enable users to determine their precise location, even in remote areas where traditional navigation methods may not be possible.

2.1.1. Parts of GNSS

The Global Navigation Satellite System (GNSS) comprises 03 main segments:

- User Segment
- Satellite Segment
- Control Segment

User Segment

The user segment is the part of the GNSS that is used by the end-user to determine their velocity position, and time. It includes all the devices and equipment that can receive GNSS signals and calculate the user's location. The devices range from smartphones to specialized receivers used in mapping and surveying applications. The user segment can receive signals from multiple satellites to provide high accuracy positioning information.

Satellite Segment

The satellite segment composes of a constellation of GNSS satellites orbiting around earth. The constellation of satellites is strategically positioned to ensure global coverage of the navigation system. The satellite segment transmit the GNSS signal to user segment. The satellites are equipped with atomic clocks that provide highly

accurate timing information.

Control Segment

The control segment is responsible for managing and controlling the satellite constellation. It comprises ground-based control stations that track and monitor the satellites. The control segment ensures the accurate positioning of the satellites, monitors their health, and manages the distribution of the GNSS signals. The control segment also upload data to the satellites, including the satellite clock and orbit data, and broadcasts it to user segment.

In summary, the user segment receives signals from the satellite segment to calculates user velocity, position and time. Control segment manages and controls the satellite constellation to ensure that every user segment receive accurate positioning information. Together, these three segments work in harmony to provide reliable and precise navigation services to users worldwide.

2.2. Working Principle of GNSS Systems

The Global Navigation Satellite System (GNSS) operates on the principle of trilateration. By measuring the distances between three known points and an object, trilateration is used to determine an object's position in space. A GNSS system uses satellites orbiting the Earth to determine the three known points.

There are three segments in the GNSS system: the space segment, the control segment, and the user segment. It consists of a constellation of satellites orbiting the Earth at a height of over 20,000 kilometers. In addition to transmitting signals with information about their location and time, these satellites also continuously send signals

containing information about their current position.

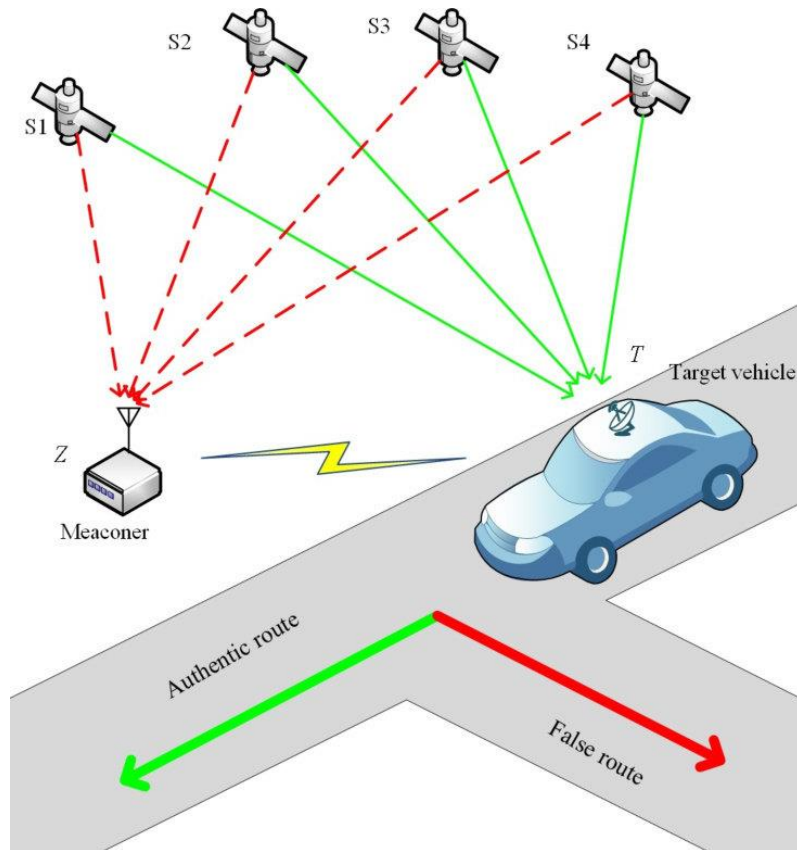


Fig. 2.3: Working Principle of GNSS Systems

Control stations are located on the ground and monitor the satellites' signals, make corrections to their orbital paths, and update the satellites' clocks. In addition to uploading navigation data to the ground stations, these stations also transmit it to the satellites for transmission to the users.

A user segment is composed of the receivers that people use for receiving and processing satellite signals. By measuring the time interval between when the satellite transmitted a signal and when the receiver received it, the receiver calculates the time delay between the transmission and reception. In order to determine the precise location on Earth, in addition to its latitude, longitude, and altitude, the receiver measures the

time delay from at least four different satellites.

Trilateration is used by the GNSS system to determine the distance between the receiver and each satellite. When the transmitting satellite sends the signal, the receiver measures the time delay between the sending and receiving of the signal. To determine the distance between a receiver and a satellite, the time delay is multiplied by the speed of light.

Once the receiver has determined its distance from at least four satellites, it can use trilateration to determine its precise location on Earth. The receiver can also determine its speed and direction of travel by comparing its position with its previous position over time.

In summary, the GNSS works on the principle of trilateration, using a network of satellites orbiting the Earth and receivers on the ground to determine precise positions on Earth. The system is used in a wide range of applications, including navigation, surveying, and mapping.

2.3. GNSS Errors

GNSS (Global Navigation Satellite Systems) works by calculating the distance from user to satellite using the time taken by the signal to travel. However, various errors can occur during this process, which can affect the accuracy of measured position. Some of the common errors that can occur in GNSS are:

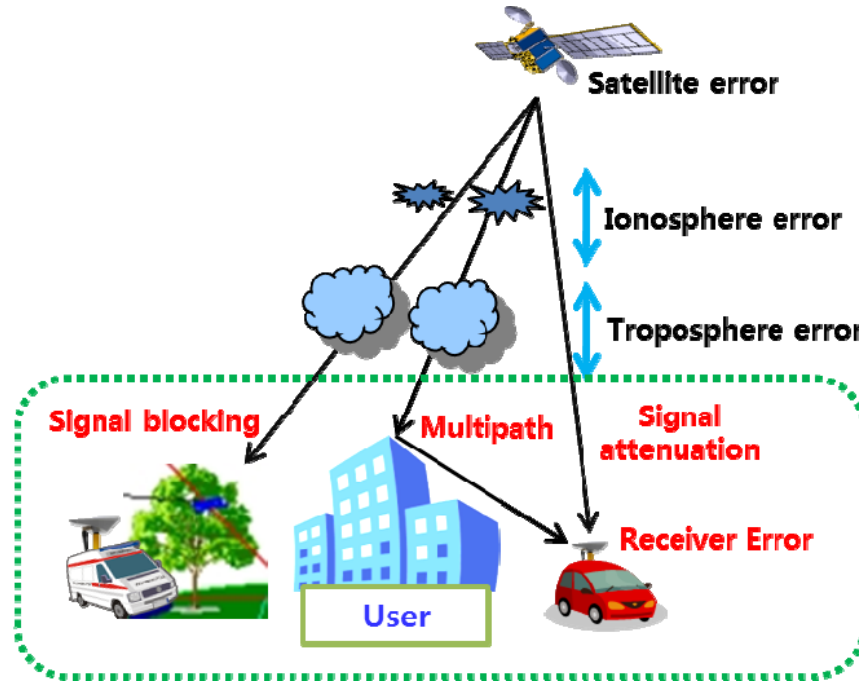


Fig. 2.4: GNSS Errors

2.3.1. Ionospheric delay

The ionosphere is layer of Earth's atmosphere which contains ionized gases. The GNSS signal travels through this layer, which causes signal delay. The amount of delay depends upon the electron density of ionosphere, frequency of the signal, and the angle of incidence of signal. This delay can cause errors in the calculated position.

2.3.2. Tropospheric delay

The troposphere is the lower most layer of the Earth's atmosphere, which have water

vapor, dust, and other particles. The GNSS signal travels through this layer, which causes a delay in signal. The amount of delay depends upon pressure, humidity, and temperature of the troposphere. This delay can cause errors in the calculated position.

2.3.3. Multipath interference

Multipath interference occurs when the GNSS signal reflects off the surface before reaching the receiver. This reflection can cause the signal to arrive at the receiver at different times, which can cause errors in the calculated position.

2.3.4. Receiver noise

The receiver can introduce noise into the GNSS signal, which can affect the accuracy of the calculated position.

2.3.5. Satellite clock errors

The GNSS satellites have atomic clock on board, which used to synchronize the signal. However, these clocks can have small errors, which can cause errors in the calculated position.

2.3.6. Satellite ephemeris errors

The ephemeris data contains information about the velocity and position of the satellite. However, this data is not always accurate, which can cause errors in the calculated position.

2.3.7. Geometric dilution of precision

The GDOP is a measure of the geometric configuration of the satellites in view.

If the satellites are clustered together, the GDOP will be high, which can cause errors in the calculated position.

To mitigate these errors, various techniques are used, such as differential GNSS, which involves comparing the signals from multiple receivers to remove the errors caused by ionospheric and tropospheric delays. Other techniques include using multiple frequencies, using advanced signal processing algorithms, and using additional sensors, such as inertial sensors, to improve precision of the measured position.

Table 2.2: GNSS error source along with their ranges

Error Source	Error Range
Satellite Clock Error	± 2 m
Satellite Orbit Error	± 2.5 m
Ionospheric Error	± 5 m
Tropospheric Error	± 0.5 m
Receiver Noise	± 0.3 m
Multipath	± 1 m

2.4. GNSS Augmentation Systems and Types

GNSS (Global Navigation Satellite Systems) augmentation system is used to improve the accuracy, reliability, and availability of GNSS positioning services. These systems work by providing additional information to the GNSS receiver, such as correction data, integrity information, or additional satellite information. This helps to mitigate errors caused by atmospheric conditions, satellite geometry, or other factors, which can affect the performance of GNSS receivers.

There are several GNSS augmentation systems in use today, including:

2.4.1. Satellite-Based Augmentation Systems (SBAS)

These systems use geostationary satellite to broadcast integrity information & correction data to GNSS receivers. Examples of SBAS systems include the Wide Area Augmentation System (WAAS) in the United States, the European Geostationary Navigation Overlay Service (EGNOS) in Europe, and the Multi-functional Satellite Augmentation System (MSAS) in Japan.

2.4.2. Ground-Based Augmentation Systems (GBAS)

This system uses network of ground base reference station to give integrity information & correction data to GNSS receivers in the vicinity of the reference stations. GBAS systems are typically used to support aircraft landing operations. An example of a GBAS system is the Local Area Augmentation System (LAAS) in the United States.

2.4.3. Precise Point Positioning (PPP)

PPP is a GNSS augmentation technique which uses additional data from reference stations to increase the precision of GNSS positioning. PPP is typically used for high-precision applications, such as surveying or geodesy.

2.4.4. Regional Navigation Satellite Systems (RNSS)

RNSS systems are similar to GNSS systems but are limited to a specific geographic region. Examples of RNSS systems include the Indian Regional Navigation Satellite System (IRNSS), the Quasi-Zenith Satellite System (QZSS) in Japan, and the BeiDou Navigation Satellite System (BDS) in China.

These GNSS augmentation systems provide critical support to GNSS users in a wide range of industries, including aviation, maritime, land surveying, and transportation. By improving the reliability and precision of GNSS positioning services, these systems help to increase safety, efficiency, and productivity across a variety of applications.

2.5. Continuously Operational Reference Station

The uses of GNSS Continuously Operating Reference Station (CORS) applications start to grow beyond geodetically control surveys and accurate orbit determination in the second half of the 1980s to include monitoring investigations as well (Lichten-Border, 1987; Rizos, 2008). CORS networks have been set up to give greater precision positioning applications. These networks have been used for many different things, including tracking crustal deformation, keeping an eye on engineering structures, figuring out tropospheric & ionospheric model, and carrying out other geosciences research in places like California, Japan, the United States, and Canada (Bock-Shimada, 1990; Chen, 1994). As a result, CORS network usage has increased, enabling positioning apps to be more exact and accurate.

The application of CORS has grown dramatically during the previous two decades, inspiring many nations to create their own networks. Several effective strategies have been put out in response to the rising demand for measurements of higher quality and in real time. Initially, post-processing was employed for numerous applications after the GPS static approach. Real-time kinematic (RTK) technology was developed in early 1990s as result of the need for real-time positioning in specific applications, such as engineering surveying, agriculture, and precision machine guidance (Rizos, 2008).

The creation of CORS networks incorporates a number of techniques suggested by

the public, academic, and private sectors. Depending on the system, these manual instructions can be categorised under a variety of headings, including technical, scientific, construction, coverage area, instrumental, conventions, standards, operation & monitoring, and much more. The NGS Guideline for Existing & New Continuously Operational Reference Station have been embraced by the majority of parties (NGS, 2020). To the growth and diversification of CORS networks, researchers and nations have also proposed exclusive rules and techniques.

Formerly, an RTK single baseline solution corrections sent from a single reference station to the roving receiver was used. RTK network solution (RTN or N RTK), which employ corrections base on the network of CORS, were instead launched as a result of further technological breakthroughs. Other network systems that use pseudo-range to give meter-level precision for long baselines include Wide Area Differential GPS (WADGPS) and the Wide Area Augmentation System (WAAS). The RTK network in contrast combines RTK with network technologies based on carrier phase measurements, enabling centimeter level precision with less reliance on the length of baseline (Rizos, 2002).

In regards to hierarchy of CORS station distribution, Chris Rizos (2008) proposed that there are three types of orders (tiers): Tier 1, which consists of the higher most international order, such as IGS stations or similar; Tier 2, which is higher national order specify by geodetic network specification; and Tier 3 which is an order define as a secondary national order. Tier 3 networks are compatible with business and industrial applications including engineering and agriculture, according to Rizos.

This categorization system offers a common method for setting up CORS networks, facilitating greater compatibility and interoperability between various nations

and enterprises. Additionally, it emphasises the value of CORS in offering highly accurate location and navigation services, which have many real-world uses in a variety of industries.

The required distances between stations for each of the three tiers were recently stated in the Guidelines of Constantly Operating Reference Station Special Publication 01. As stated in the recommendation, The spacing between Tier 1 stations, which are the highest international order stations like IGS stations or comparable, should be between 500 and 1500 kilometres. When it comes to geodetic network specifications, Tier 2 stations, which are the highest national order stations, should be spaced 80 to 500 km apart, while Tier 3 stations, which are the secondary national order (state order) stations, should be spaced 20 to 80 km away.

It should be mentioned that the baseline lengths of the reference stations vary on various elements, including the size of the nation, the distribution of its people, and the level of accuracy that is sought. Active Control Stations (ACS) are arranged in dense networks to attain high levels of accuracy. In real time kinematic RTK applications, both for network base solution, single base and the length of the baselines is a crucial component in determining the quality of positioning. Researchers like Chen et al. (2001) and Li et al. (2014) have recommended that in mid-latitude regions that are more susceptible to strong ionospheric impacts, the baseline length be restricted to 50 km or less.

To cover such a huge region with stations spaced at 50 km intervals would need a significant commitment of resources, therefore reaching the required density of reference stations in big countries like China or Australia can be difficult. Researchers

have been working on innovative techniques to overcome this difficulty and raise the inter stations distances to at least 100 km while preserving centimeter-level positioning accuracy (Li et al., 2010; Tang et al., 2010; Silva et al., 2020).

Critical elements of network design include the number of reference station & their placements, which are determined by factors including the size of the nation, population density, and strategic initiatives. Nonetheless, the limitations caused by baseline length have been lessened thanks to the advancement of GNSS technology. Newer antennas and receivers can now receive signals from a variety of satellite constellations, including Galileo, GPS, BeiDou and GLONASS which can gradually fill in any gaps in the network coverage. Globally Galileo, GPS, BeiDou and GLONASS and locally, QZSS and IRNSS. Figure 2.5 depicts the number of satellites that were visible in the sky over the Institute of Space Technology, Pakistan, on August 24, 2022.

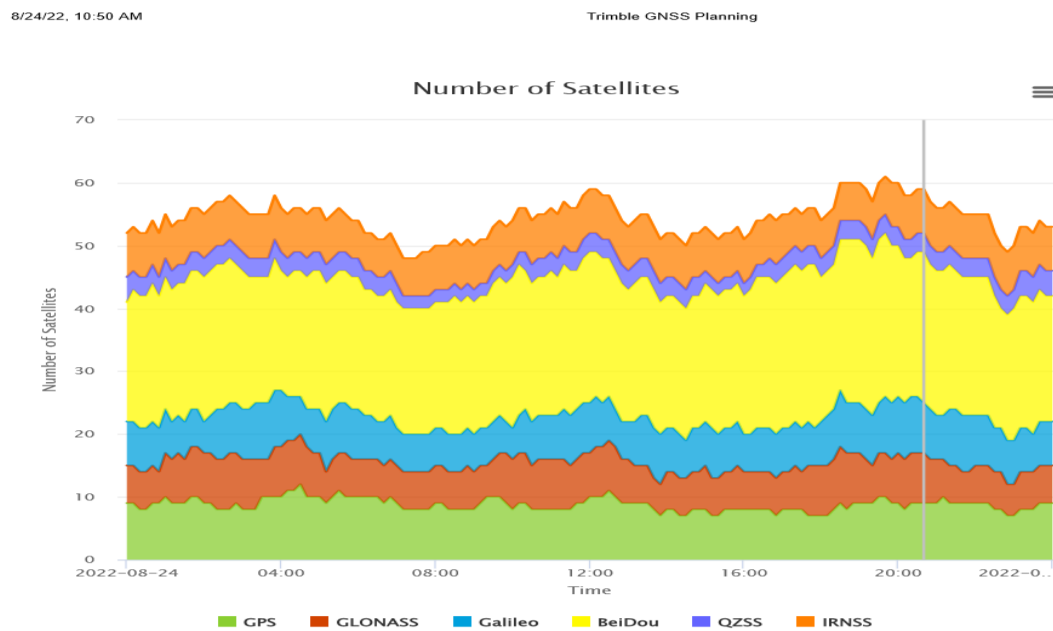


Fig. 2.5: Stacked area chart showing the number of visible satellites above Institute of Space Technology, Islamabad Pakistan on the date 24th August 2022 (Trimble GNSS Planning Online)

Hein (2020) asserts that having numerous satellites and frequencies accessible has some drawbacks, including compatibility and interoperability concerns. The quality and dependability of Real-Time Kinematic (RTK) Positioning, however, can be improved by adding additional constellations in addition to GPS because of improved ionospheric modelling, a higher likelihood that they will be available, and increased resistance to interference and multipath effects that result from using multiple frequencies.

The use of numerous constellations may have advantages, but there are compatibility and interoperability issues that must be resolved. In order to achieve the seamless integration of various constellations for high-precision locating applications, Hein (2020) stressed the significance of finding solutions to these problems. Furthermore, the design of GNSS CORS is impacted by the use of various satellite constellations and frequencies. A network's reference stations' placement and quantity are determined by its coverage region, population distribution, and tactical initiatives. With enhanced tropospheric models and ambiguity resolution techniques, researchers have been looking into ways to extend inter-station distance while preserving centimeter-level precision (Li et al., 2010; Tang et al., 2010; Silva et al., 2020).

Overall, quality and dependability of high-precision RTK location can be improved by integrating numerous constellations into the GNSS CORS design. To ensure the seamless integration of various constellations, compatibility and interoperability challenges must be resolved.

Table 2.3: Illustration of Existing Augmentation Systems

Sr. No.	Name of Augmentation System	Origin of country	Coverage Area	Signal in Space Range Error	Ionospheric Correction Accuracy
1	Wide Area Augmentation System (WAAS)	United States of America	North America	0.954m	0.505m
2	European Geostationary Navigation Overlay Services (EGNOS)	European Union	Europe, Africa and Partially Canada	0.645m	0.491m
3	GPS Aided GEO Augmentation Navigation (GAGAN)	India	Indian region, Africa and Australia	0.795m	0.858m
4	Multi-functional Satellite Augmentation System (MSAS)	Japan	Japan region	1.931m	1.325m
5	System for Differential Corrections and Monitoring (SDCM)	Russia	Russian region and partial Europe	0.650m	0.523m

2.6. CORS Design Parameters

GNSS networks are designed according to the three main parameters set forth in the Guide for CORS Publication (ICSM - PCG, 2014). In the research of this thesis, it is the first and third parameters that play a significant role in creating a coverage plan. Moreover, it is necessary to lighten the extra directions that are essential to successful system's installation. Following are the key points to consider:

1. Length of the baseline
2. The reference frame should be linked globally and/or nationally
3. The service is affected by a malfunction at a station

2.6.1. Effect of Baseline length

In addition to being affected by numerous factors, determining the maximum distance between stations can have a significant impact on performance. Disturbances in the orbit and in the atmosphere are important factors that influence base-rover separation. In general, this correlation holds true, but is particularly important when dealing with real-time positioning. Due to the nature of RTK technique, short observations are usually made, resolution of ambiguity is needed quicker and more reliably, and broadcasting of ephemeris data is utilized. Baseline length constraints are applied in a variety of ways cautiously considered from them to provide a fast fix results, while ensuring a high degree of accuracy and validation. There are three levels of separation between the stations, as determined by the ICSM and PCG (2014): First-order from 500 till 1500 Km.

- Distances between 80 and 500 kilometers of second order
- Distances between 20 and 80 kilometers of third order

To achieve an effective design, it is imperative to specify and prioritize the number and spread of stations. Developing scenarios encompassing strategic perspectives can serve as a starting point for this process. There are advantages and disadvantages to each scenario, and the objective is to choose the best available option which will facilitate post-installation future development & functioning. In this study, we take into account

aspects and scenarios specific to the Pakistan Region. A number of factors need to be taken into consideration, including:

- Settlements and large cities
- Other strategic projects, such as dams and oil fields
- Plan that provides the best coverage
- Whether there is a single CORS system or a network of CORS systems

2.6.2. Malfunction Effects

In the condition of second order with Real Time Kinematics network systems, it is highly recommended to check the disorder of the network before launching the final system. Tests should be conducted on both the territorial and the whole network to determine the impact of the disruption caused by a station. In order to predict the effects of disruption, simulations and manual investigations can be used. It is through these results that weaknesses can be improved and a more reliable system can be achieved. It will assist in determining the likelihood of stability of service if power supply, alternative station, hardware, connection services and software are used. However, the type of system or generation also plays a key role in determining the severity of the malfunction. The solution is affected by a unique factor regardless of whether the system is a single or a CORS network.

2.7. CORS Establishment:

In this section, all the necessary requirements are outlined for the application of the design and the establishment of the network. The exact location must be determined once the approximate location has been determined in the planning section. In The designated sites must meet a range of requirements, including security, power, stability, satellite visibility, power, communications, data standards, data sharing, and data management, as well as topography and weather conditions. The given instructions are a combination of NGS given Guidelines for Existing and New CORS Network (NGS, 2020), UNAVCO Guidelines for CORS Network (ICSM – PCG, 2014), and Guideline for CORS Special Publication 1. Given guidelines provide the details about the steps involved in the establishment, which are discussed in this thesis.

2.7.1. Signal quality assessment

Before constructing or mounting an antenna, it is extremely necessary to check the signals of receive. This test involves controlling the sky visibility using an above the horizon of elevation angle, carefully selecting an area that takes into account the occurrence of multipath due to the manmade and natural and object, & minimizing interference in signal. When the object such as solar power systems, antennas, fences and objects are installed, a minimum length of 20m must be observed, and a minimum angle of elevation 5° must be observed. GNSS links used to transfer data can themselves be sources of radiofrequency interference (RFI). RFI factors increase when the interference frequency is harmonic with the signal frequency GNSS. The presence of other interfering sources. In addition to mobile phones transmitter, radio, microwave networks, television, transformer & power lines, GNSS signals are transmitted through several different methods.

In this case observation can be performed using sensors, provided the view is clear, the required sky visibility is maintained, and obstacles are minimized. An antenna can be replaced with cameras to take panoramic images of the environment (360 degrees). By doing so, we can determine the elevation angles of obstacles (figure 6). Additionally, laser scanners can check the obstruction's distance from antenna & elevation angle using their laser beams. An assessment software such as TEQC is used to perform the signal quality valuation step. It is recommended to record at 1Hz, use an elevation mask of 0° , and to not smooth the pseudorange for at least two days (48 hours).

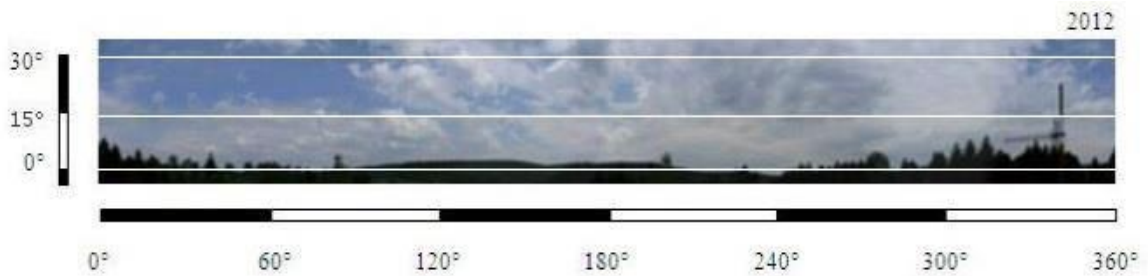


Fig. 2.6: View of the obstacles surrounding the CORS site as visualized by a 360° camera (from the Hungarian GNSSnet.hu)

2.7.2. Permission

Getting permission from the responsible owners, which could include private or governmental entities, is required before to the construction & site investigation. It is necessary to take geophysical samples of the soil for the construction of the monument under government regulations. In addition, existing infrastructure and the environment must be considered.

2.7.3. Site Access & Security

An efficient maintenance of the system is important since the system is not only established but also sustainably maintained. The station's location should be easily accessible and easily reachable by authorized personnel during adverse weather conditions. The CORS site and equipment should also be protected from weather, theft, animal, insects, destruction, & lightning. Site security is a necessity due to factors such as future dependence on the station for geodetic work and facilities as well as the expenses associated with establishing the station.

2.7.4. Site stability

Monuments and antenna mounting devices are essential to the stability of structure of antenna's foundation. Depending on CORS order, the foundation is laid. To achieve the highest degree of stability, the first and second orders should be secured to deep rocks with extreme high stability. It is important that the geodetic reference frame has a high level of stability because this is used whenever recording dataset regarding motion of tectonic plates, establishing the national geodetic reference frame tidal forces. The best location for ensuring consistency and longevity is one that exposes the bedrock. For foundations exceeding 4 meters deep, geotechnical testing of the soil is mandatory, as it may be necessary to use large concrete foundations.

As the third order, the antennas are existing & fixed concrete structure, taking advantage of their power, communication, and security. Because wind and thermal expansion are more significant when structures are higher than ten meters (two-story building), it is recommended that buildings no higher than ten meters be used. NGS guidelines (NGS, 2020) provide more details regarding this topic. Additionally, because

buildings are prone to damage during earthquakes, monuments should be used as opposed to existing structures.

There are two primary types of UNAVCO Braced Monuments: deep and shallow (drilled and non-drilled). There are advantages to shallow braced monuments over concrete pillar monuments. They are designed as equivalent classes. There are four or five legs of stainless steel with 01 inch dia each that comprise a shallow drilled braced monument. In order for the drilled type to be effective, it must be deployed on deep bedrocks up to approximately 5 and 6 feet deep (or on a surface where there is bedrock under 0.5 meters). Weather impact and fractures are not present in an ideal bedrock. There are three networks using this method: PANGA, BARGEN, and PBO. There are no differences in the type of still legs and dimensions between shallow non-drilled braced monument and shallow drilling monuments. There are two types of pounded monuments: those whose legs are pounded into loose ground, and those whose holes are bore by hand. Figure 7 and table 2 The UNAVCO website displays comparisons between some types of monumentation.

Antenna mounting structures of three types by UNAVCO:



Fig. 2.9: Braced Monument



Fig. 2.8: Concrete Pillar



Fig. 2.7: Building Mounting

Antenna monuments should meet the following characteristics according to ICSM and PCG (2014):

- Stability over the short, mid, and long terms
- A minimal amount of multipath
- To minimize obstructions, the height must be sufficient
- In Tiers 1 and 2, true verticality must be within 1 mm, while in Tier 3, it must be within 5 mm
- An easy-to-assemble, easy-to-install, and easy-to-maintain design
- Easy to maintain
- Easily corroded, eroded and buried
- Suitable for supporting antenna mass
- Design that prevents tampering

Stability standards must be met for concrete monuments by engineers. A monument must have a lower top diameter or width than an antenna to minimize multipath effects. The recommended height can also be between 20 and 25 cm for orders first and second, respectively

There are two options for 1.2 and 1.7 meters, which will reduce the effects of multipath, wind load, and expanding. Three-order stations require stainless steel that has specific properties. GNSS carrier phase wavelengths (19 or 24 centimeters) should not be multiplied by antenna heights on structures with metal roofs or other reflective surfaces (ICSM - PCG, 2014).

2.7.5. Time and cost management

It is possible to consider time and capital as complementary and alterable elements, i.e., good time management can lead to a system of economic growth and powerful finance can facilitate a quick execution process. Establishing a CORS network requires the implementation of a good management plan, similar to engineering projects. The cost of the system must be considered along with all the necessary equipment and services. An exhaustive review of all expenditure sources is necessary to obtain good understanding. An estimation of the minimum known costs can be used to determine the costs for one station's establishment, but all costs may change in the future. Hardware and software costs, services, labor costs, fuel costs, maintenance, managing, and security costs, and construction costs can be generalized for each station.

Finance, availability, and reliability of the system are all impacted by time, as a critical parameter. More system needs to be completed in a timely manner, additional personnel, equipment, and other resources will be required. As a result, there will be higher expenses associated with the project. The system's reliability also depends on its ability to manage time, to ensure fast maintenance, and to provide excellent customer service.

2.7.6. Maintaining control and management

In the following stages, users and management become able to access the system after it has been established. Remote control, site security, and regular field checks are all parts of such management. It is essential to conduct regular field visits to ensure that the network runs smoothly and reliably. In addition, it is necessary to inspect the GNSS device, the power supply, the communication system, and the weather measuring devices to ensure that they are working properly and that the environmental conditions have not changed.

Using these visit bases, one can determine when equipment needs to be replaced and how to reduce multipath and radio frequency interference effects on the site. There are many ways to maintain site security. An example is to use a secure location or building to house the GNSS instrument, to use fences (carefully designed to avoid multipath), to provide fire protection, to secure the power supply, and to limit access.

It is very important to have a remote control during the process. An internet connection, telecommunication, computers, websites, and software is usually used in the central monitoring station (Master Station). By using computers and surveillance, this can provide robust dynamic analysis dashboards for remote monitoring. Distant monitoring has the advantage of decreasing the number of field checks and management costs associated with that. In addition, the website makes it possible for the customers to access numerous data analyses through tables, charts, maps, and plots through an interactive, smart, and user-friendly interface.

2.7.7. Data standards and sharing

GNSS is a collaborative rather than an individual system, as evidenced by its instrument and its working mechanism. While using the stand-alone GNSS technique, data are shared between several satellites, while using the RTK technique, data are shared with other instruments on the ground, as there are multiple links. However, there is also an important issue of interoperability due to the interconnection between the various manufacturers of instruments, as well as between neighboring countries. Data formats, sharing protocols, and communication links must be standardized to meet these requirements. Globally, NTRIP, RINEX, RTCM, & NMEA are the primary data standards used today (Heo, 2009).

To manage network communication in between master station and the auxiliary stations, which are in continuous communication, a network management system is required. It is possible to give the results alongside the website exclusively designed for network by using specially designed software that maintains a smooth data transmission.

2.8. Other Important Aspects

2.8.1. Environmental and topographic considerations

GNSS single baseline and network measurements are affected by the terrain type and geographical location. Consequently, the tropospheric error would vary due to the variation in height between stations. Environmental measures can also be affected by topography and location. A temperature reading, a barometric pressure reading, and a humidity reading are the most important factors. Due to the above reasons, it is necessary to study the area from a geographical standpoint, create a map of the area based on cartographic data, and install environmental gauges in order to monitor the changes. Telecommunication coverage and data transmission are also challenging in the Islamabad Region due to the large variation in height. The establishment of reliable communication and alternative methods of communication in case of a shortage requires an extensive study.

Islamabad, Pakistan, is depicted on the topographical maps. Throughout the country and in particular in the Islamabad Region, there is a large variation in elevation.



Fig. 2.10: Topographical map of Pakistan.

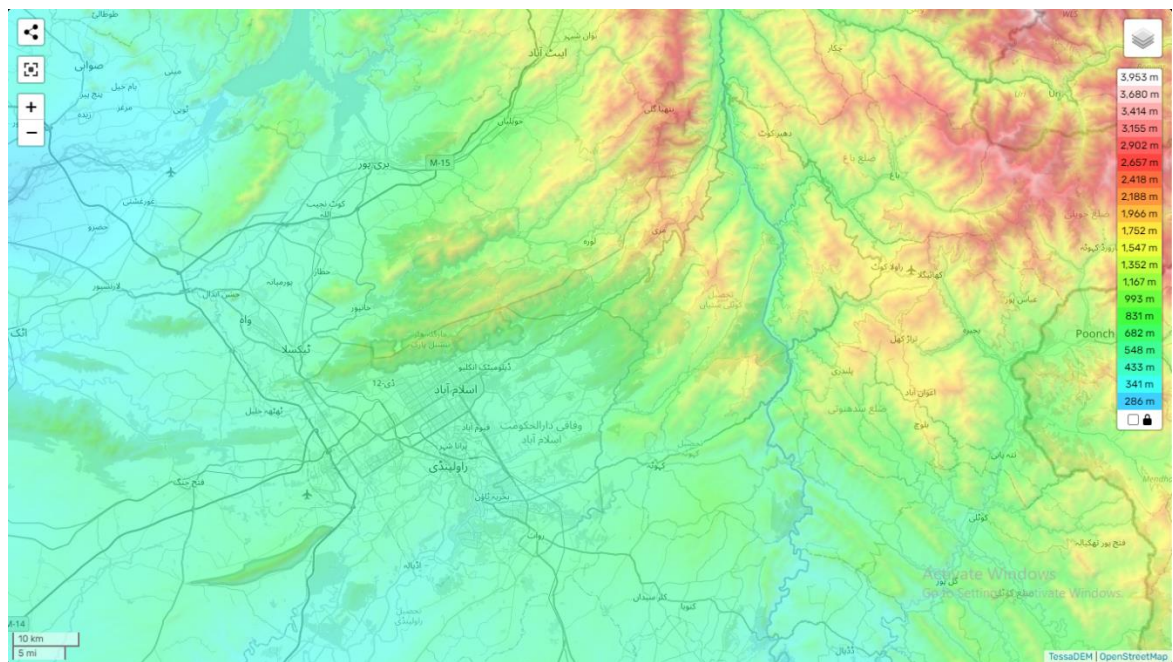


Fig. 2.11: Topographical map of the Islamabad Region

3. Methodology

3.1. Data Collection and Experimentation

For the course of this experiment, multiple datasets were collected in post processing to develop, test and verify the algorithm of the continuously operating reference stations. For the collection of datasets, the u-blox zed F9P (dual frequency) and u-blox C94 M8P (single frequency) receivers were used. The dataset collection frequency was kept at 1 Hz and Football ground near IST boys hostel was selected as the site of data collection. The Zed F9P is a dual frequency multi constellation GNSS receiver that supports GPS, BDS, GLONASS, Galileo, QZSS, & SBAS. In contrast, the C94 M8P is a single frequency GNSS receiver that supports GPS, GLONASS, & QZSS constellations simultaneously.

During this experiment, the Zed F9P receiver was made the base station by surveying its true coordinates while the C94-M8P receiver was acting as the rover receiver that received the corrections from the base station and computed its accurate position, velocity and time with the help of the corrections received from the base station. Although the CORS network can be of multiple base stations but for the sake of this experiment and for the proof of concept only one base and one rover was selected. However, the working principle is same for both single base and multiple base stations.

The base station can be set in two different methods, i.e. either the true position of the base should be known from third party sources or it is surveyed using the receiver's built-in survey-in feature. Fortunately, the u-blox receivers and the u-center software comes with an easy to use user interface along with various built-in features that makes configuring the base station as well as the rover station easy as compared to setting it manually. In order to set the base station, the receiver (i.e. the u-blox zed F9P) was

connected to the comm. port 5 of the host PC. As soon as the receiver was plugged to the comm. port, the operating system (windows 10) installed in the host laptop automatically detected and installed the drivers of the receiver and the receiver was operational in a while. However, sometimes this may not be the case depending upon the operating system installed in the host computer, availability of the internet connection, and environment conditions. Under such conditions, the drivers can be downloaded from the u-blox main website and installed using the device manager option in the windows.

3.1.1. Connecting the Receiver and Installing Drivers

The following are the main steps to ensure that the connected receiver is properly installed and operational:

1. Go to device manager

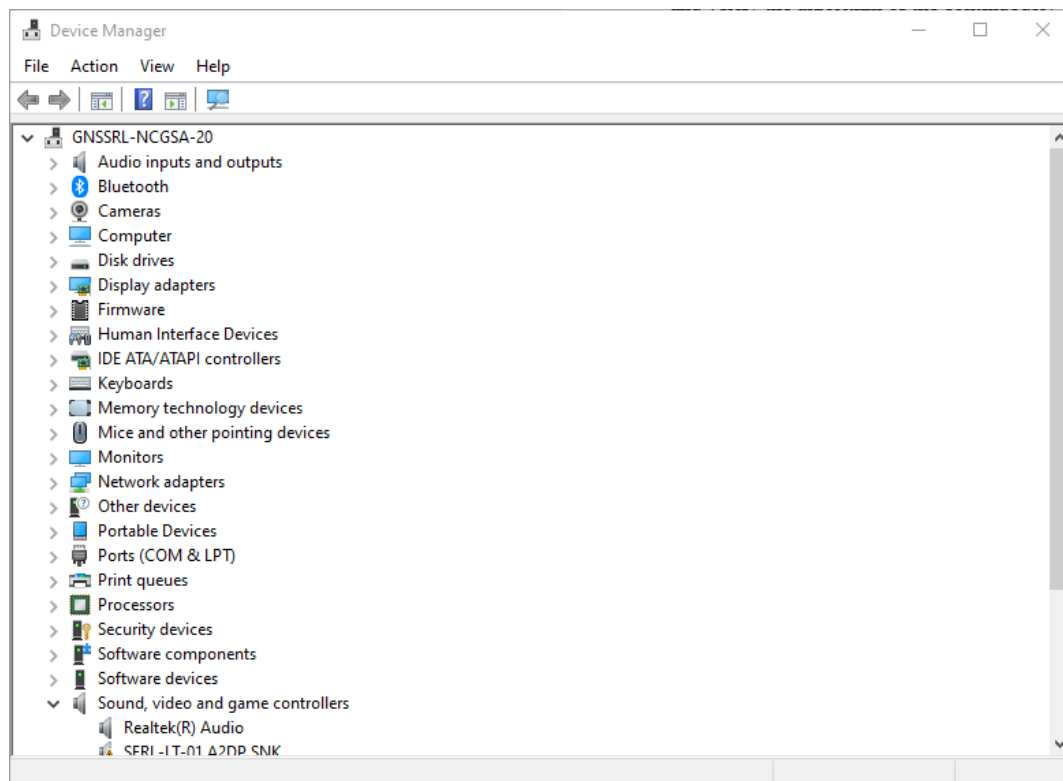


Fig. 3.1: Device Manager

2. Click on the Ports (COM & LPT) and a drop down menu will appear.
3. Now confirm whether the receiver is properly installed or not.
4. If the drivers of the receiver are missing, then there will be a small yellow triangle on the receiver icon which indicates that the drivers need to be installed again.
5. Under such conditions, right click on the icon with the yellow triangle and a drop down menu will appear.
6. Click on update drivers and a new window will open with two different options, i.e. search automatically and install manually.
7. Since the system was unable to install the drivers on its own therefore, there is a higher chance that the search automatically feature will fail. The manual installation is a much better option under such conditions.
8. In order to install the drivers manually, they have to be downloaded from the given link and extracted to the local drive of the host computer first.
9. Now click on the install driver manually option and browse for the folder with the downloaded drivers for the receiver and click ok.
10. It may take a moment to verify and install the drivers and the receiver will start working.
11. A blue and a green light blinking on the Zed F9P board confirms that the host PC has properly identified the connected receiver.

3.1.2. Setting the U-Blox ZED F9P as Base Station

Once the receiver is connected the host computer and it is working fine, the next step is to set it as a base station in order to have a reference with exactly known coordinates. This can be done by interfacing and controlling the receiver from a software called u-

center. U-Center is a u-blox proprietary software that is used for interfacing and configuring their receivers directly from the software. U-Center can be downloaded from the given link [\(\)](#) and install on the computer. After installing the u-center, the following steps should be followed in order to configure the base station with the u-blox zed F9P receiver.

1. Choose the right comm. Port to which the receiver is connected. During this experiment the receiver was connected to the comm. Port 5 (USB 2.0).
2. After selecting the right comm. Port of the receiver the NMEA messages will start to appear and the main window of the software will populate with different messages including SNR, sky plot, receiver position information, fix etc.
3. Now go to view > messages view > and right click on the NMEA messages to and select disable child messages.
4. Disabling the NMEA child messages helps saving the baud rate and prevent the loss of necessary data.
5. Now scroll down and open UBX > CFG > TMODE3.
6. Inside the TMODE3 option, there are two options that can be selected from the drop down i.e. set the receiver exact coordinates previously known from third party sources or allow the receiver to calculate its true coordinates with minimum error using the built-in survey-in feature.
7. During this experiment, the receiver was set to survey-in mode to find its true coordinates that are accurate up to 1 meter and the minimum duration of observation was set to 600 seconds.
8. After properly configuring the base station the positioning mode on the right side

of the main u-center screen will shift from 3D/DGNSS to TIMED/DGNSS.

9. This indicates that the receiver has found its true coordinates and it is now set as a base station.
10. The next step is to configure the RTCM MSM messages that the base station will send to the rover receiver nearby.
11. Go to ubx > CFG > PRT and set the protocol in to none and protocol out to ubx + NMEA + RTCM 3.
12. Go to MSG from the messages view inside the CFG menu and configure the following messages on UART 1.
13. Now go to ubx > RXM and enable both the RAWX and SFRBX messages.
14. For the last step save the receiver configurations from the receiver option in the u-center as shown in the figure and disconnect the receiver from the u-center.

Table 3.1: RTMC MSM Messages and Their Frequencies on UART1

Message Type	Frequency
RTCM 3.3 1005	1
RTCM 3.3 1074	1
RTCM 3.3 1077	1
RTCM 3.3 1084	1
RTCM 3.3 1087	1
RTCM 3.3 1094	1
RTCM 3.3 1097	1

RTCM 3.3 1124	1
RTCM 3.3 1127	1
RTCM 3.3 1230	1
RTCM 3.3 4072.0	1
RTCM 3.3 4072.1	1

3.1.3. Configuring the C94 M8P as a Rover Receiver

After properly configuring the base station and setting it to sending the messages to the rover receiver, the next step is to set the rover to receive the messages from the base and apply these correction message to improve its positioning accuracy. In order to set the rover receiver here are the steps to follow:

1. Connect the rover receiver to the host PC.
2. Go to view > messages view
3. Right click on the NMEAS messages and disable the child messages.
4. Go to UBX > CFG > PRT
5. Set the port to UART1 and protocol out to none and protocol in to ubx + RTCM 3 + NMEA.
6. Click send to send the command to the rover receiver.
7. Go to ubx > rxm and enable the RAWX and SFRBX messages.
8. Go to receiver and save the configurations of the rover and then disconnect it from the u-center.

9. The receiver will now act as a rover and will start applying the correction messages once it receives them from the base station.

3.2. Communication Between the Rover and the Base Stations

The heart of the CORS network is the communication channel between the rover receiver and the base receiver through which it receives its corrections and improve its overall accuracy. This can be done through numerous methods including RF communication using radio link, via Bluetooth, or through the internet server also known as the Network Transport of RTCM via Internet Protocol (NTRIP). Since the former communication means are limited to distance therefore, the internet communication via NTRIP is the most suitable option as it allows the receivers with a working internet connection at any distance from the base to get the RTCM MSM messages. When it comes to establishing the communication between the base and the rover via an internet connection, there are two ways connecting them:

- Through a local server.
- Global server using NTRIP caster/client setup.

3.2.1. Connecting the Receivers Through Local Server

When connecting and communicating the two devices via a local server, both have to be on the same internet connection. A local server connection can be established using the following steps:

1. Find the IP address of the internet connection to which the host PC is connected.
2. Run the STRSVR module of the RTKLIB, an open source C++ library for GNSS processing.

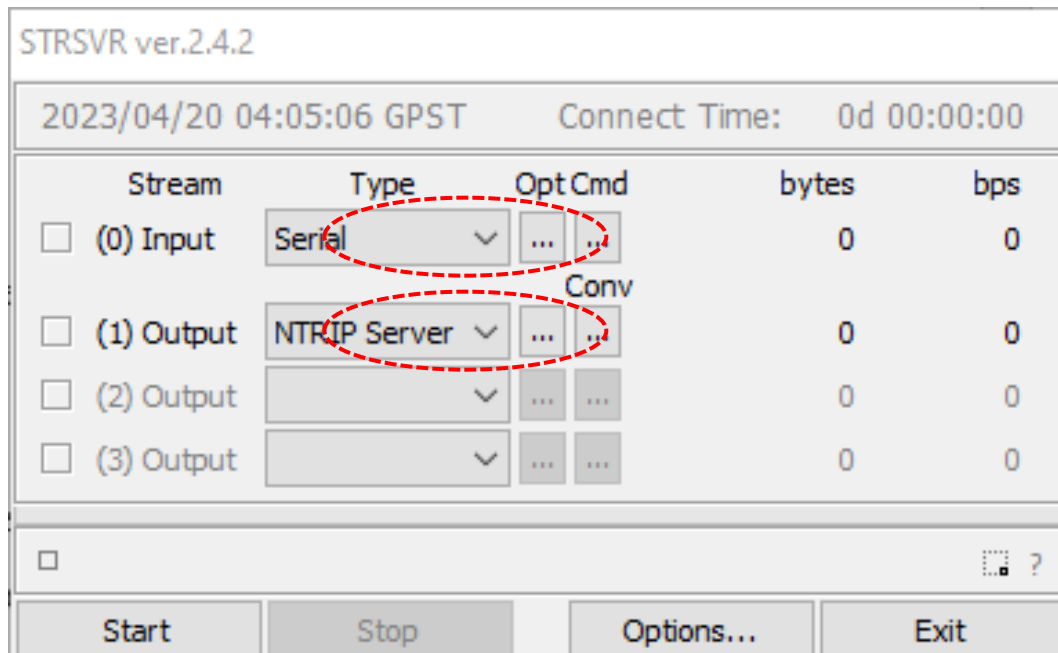


Fig. 3.2: STRSVR module of the RTKLIB

3. When the window of the STRSVR appears, set the input type to serial.
4. Click on the three dots option, next to the serial and choose the comm. Port where the GNSS receiver is connected as shown in the Fig. 3.2.

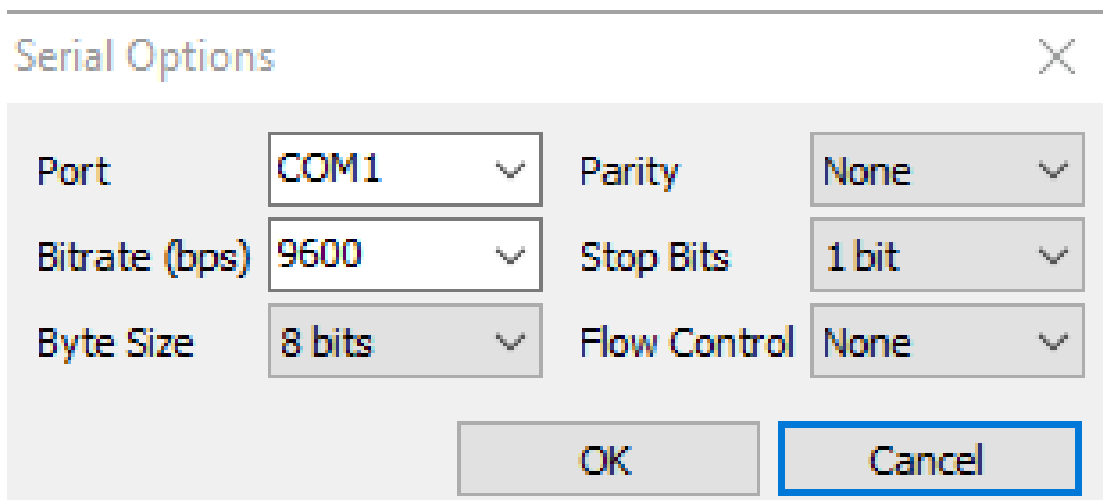
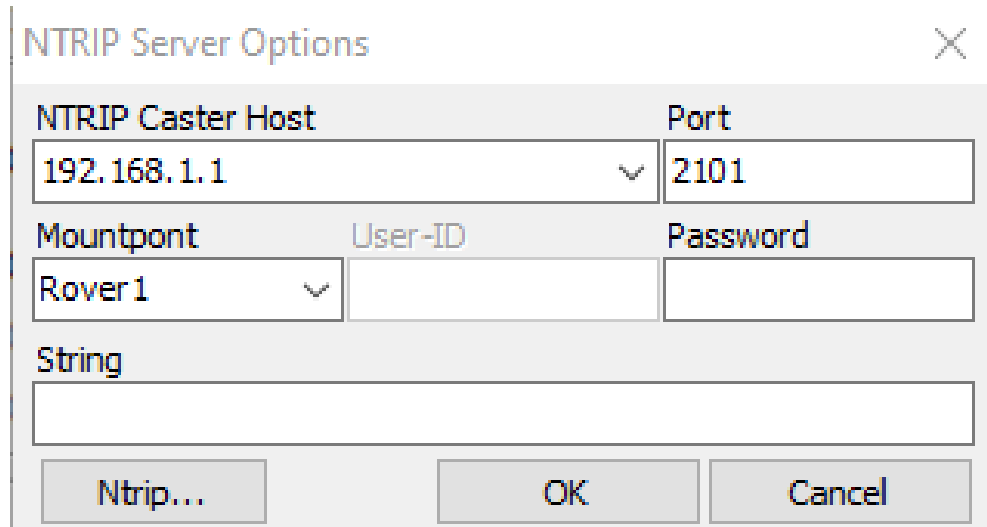


Fig. 3.3: Port Selection Window.

5. Next set the output of the receiver to NTRIP server as shown in the figure above

and click on the three dots next to it.



NTRIP Caster Host		Port
192.168.1.1		2101
Mountpoint	User-ID	Password
Rover 1		
String		

Ntrip... OK Cancel

Fig. 3.4: Configuration of NTRIP Caster

6. Configure the NTRIP using your IP address as NTRIP caster Host, and set the port to 2101.
7. Name your Mount Point or in other words the station name.
8. Click on the ok button and start the STRSVR.

Once the base station is successfully connected, it will start getting the data and send it to the NTRIP server. For the rover receiver, run the STRSVR and the above steps are followed except this time the input is taken from the NTRIP server (use same configuration) and the output is set to comm. Port to which the rover is connected. The following are the steps to configure the rover station for receiving the RTCM MSM messages from the NTRIP server:

1. Run STRSVR

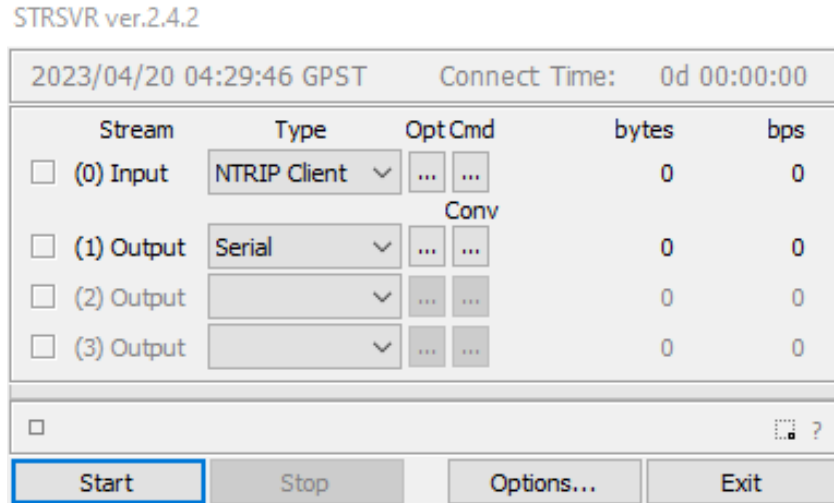


Fig. 3.5: STRSVR Start Window

2. Set the input to the NTRIP client and use the exact same options.
3. Set the output type to serial and select the comm. Port to which the rover is connected.
4. Click on the start button to start the rover receiver getting input from the local server

3.2.2. Connecting the Receivers Through a Global NTRIP Server

Although establishing the communication between the rover and the base through a local server is pretty straightforward, the main problem in doing so is that both the receivers must be on the same internet connection otherwise they won't be able to communicate on the same IP address. To communicate distant receivers with each other, there has to be a communication channel that is independent of the internet connection to which both devices are connected. Under such conditions, an NTRIP caster/server is used as a medium of communication instead of the IP address. This can be configured using the address of the NTRIP server in the STRSVR as shown in the figure There are multiple NTRIP servers available worldwide and some of them are free of cost. For the course of this experiment Emlid Caster was used as a global NTRIP server.

4. Results & Discussion

During this research multiple datasets were collected at different times and environment conditions. The collected datasets were in binary format (ubx) which were converted to receiver independent exchange format (RINEX) using the RTKCONV module of the RTKLIB. It should be kept in mind that the datasets can be converted with a variety of software packages. Sometimes, the RTKLIB might not be able to produce the Navigation file for the tracked satellites which converting the datasets. Therefore, it is better to use Emlid Studio if such a problem occurs.

4.1. Base Station Position Calculation

The datasets recorded with the base station are processed first and their position was calculated using the single point positioning. Furthermore, the observation file was uploaded to an open source position correction service called CRS-PPP to verify the true position of the base station along with the error in coordinates (in East North Up frame).

The following are the results obtained after the calculation of the position of the base station:

During the data collection with the base station the receiver initially tracked a total of 10 satellites. Furthermore, the geometric dilution of precision recorded to be 1.8. In addition to the GDOP value, the position dilution of precision (PDOP), Horizontal dilution of precision (HDOP) and vertical dilution of precision (VDOP) were recorded to be 1.6, 0.8, and 1.4 respectively. The following figure illustrates the sky plot of satellite tracked by the base station.

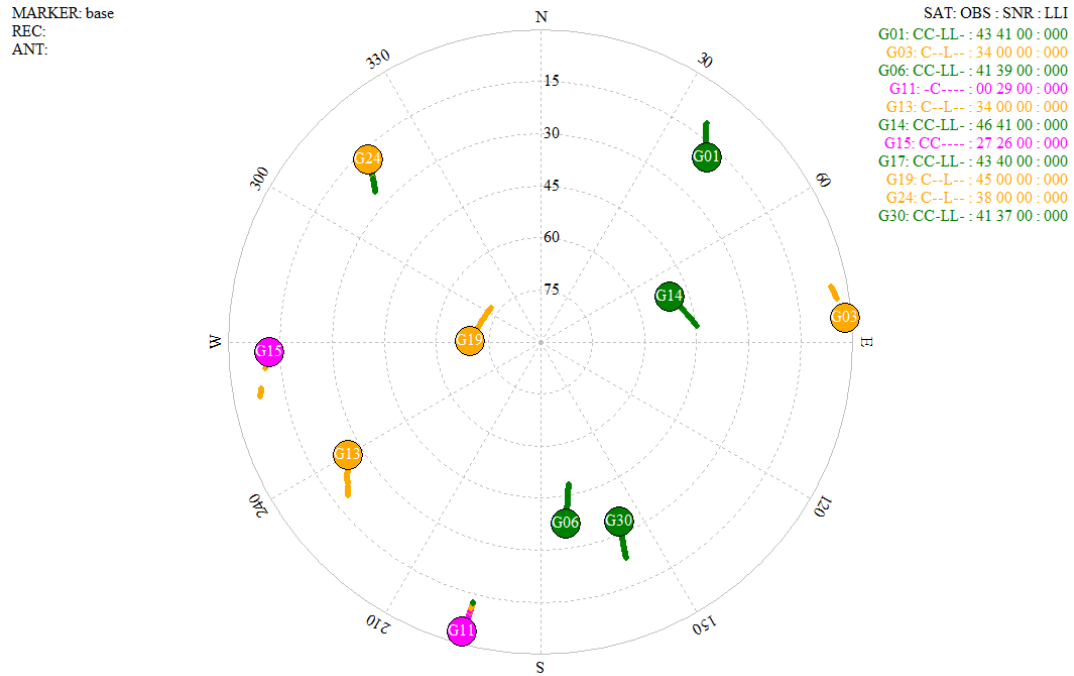


Fig. 4.1: Sky plot of satellite tracked by the base station.

Apart from that, the figure 4.2 provided displays the signal to noise ratio (SNR) of the satellites being tracked, including an analysis of multipath, as well as SNR in relation to the elevation angle of each satellite as detected by the receiver.

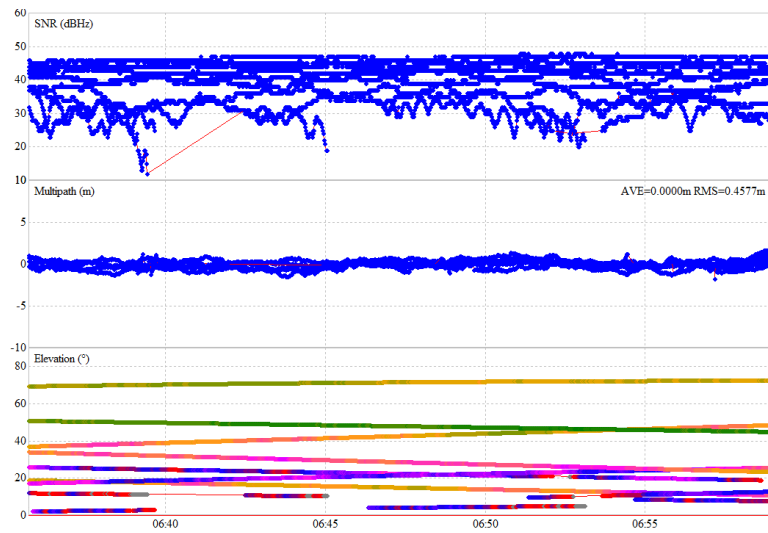


Fig. 4.2: SNR, Multipath and Elevation Angle of detected Satellites.

The position of the base station was determined using the collected datasets, and the results obtained are represented in the following figure. From the figure, it can be observed that the calculated position of the base station resulted in a standard deviation of 1.3 m in the north coordinate, 1.1 m in the east coordinate, and 3.1 m in the up coordinate.

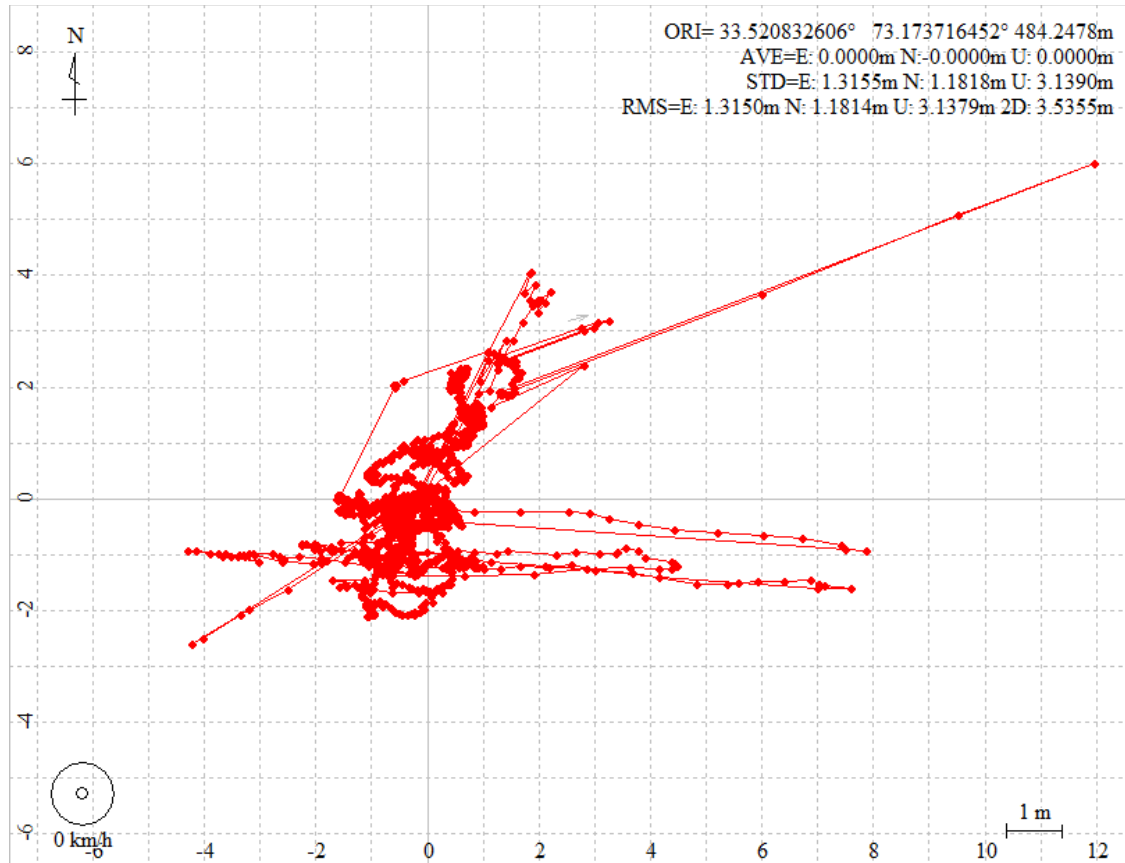


Fig. 4.3: Position of Base Station

4.2. Base Station Position Correction With CRS-PPP

Since the position of the base station needs to be highly accurate therefore, it was surveyed using the Canadian Precise Point Positioning Service (CRS PPP) to calculate the true position of the base station along with other necessary information. The following are the results obtained from the CRS-PPP service:



CSRS-PPP 3.54.2 (2022-11-10)



base1_1_.23O

Data Start	Data End	Duration of Observations
2023-04-06 06:35:43.00	2023-04-06 06:59:01.00	0:23:18
Processing Time		Product Type
05:27:04 UTC 2023/04/20		NRCan Rapid
Observations	Frequency	Mode
Phase and Code	Double	Static
Elevation Cut-Off	Rejected Epochs	Fixed Ambiguities
7.5 degrees	0.00 %	0.00 %
		Estimation Steps
		1.00 sec
Antenna Model	APC to ARP	ARP to Marker
Unknown	Unknown	H:0.000m / E:0.000m / N:0.000m

(APC = antenna phase center; ARP = antenna reference point)

Estimated Position for base1_1_.23O

	Latitude (+n)	Longitude (+e)	Ell. Height
ITRF20 (2023.3)	33° 31' 14.86700"	73° 10' 25.33152"	483.563 m
Sigmas(95%)	0.116 m	0.327 m	0.289 m
A priori*	33° 31' 15.04310"	73° 10' 25.38940"	490.804 m
Estimated – A priori	-5.426 m	-1.494 m	-7.241 m

Fig. 4.4: Results obtained from CRS-PPP

The true position calculated by the CRS-PPP service and the following figures represents the variation in Lat, Long and height of the base receiver along with its corrections.

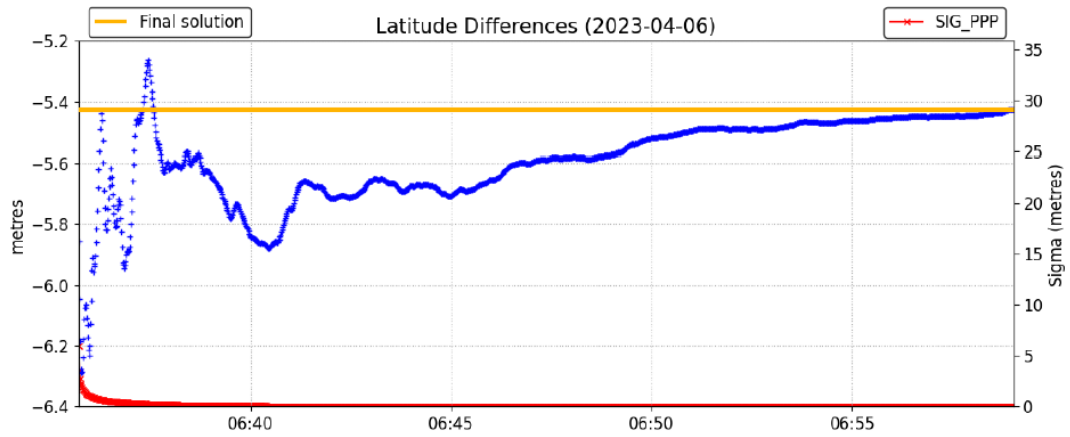


Fig. 4.5: Latitude Variation of Base Station.

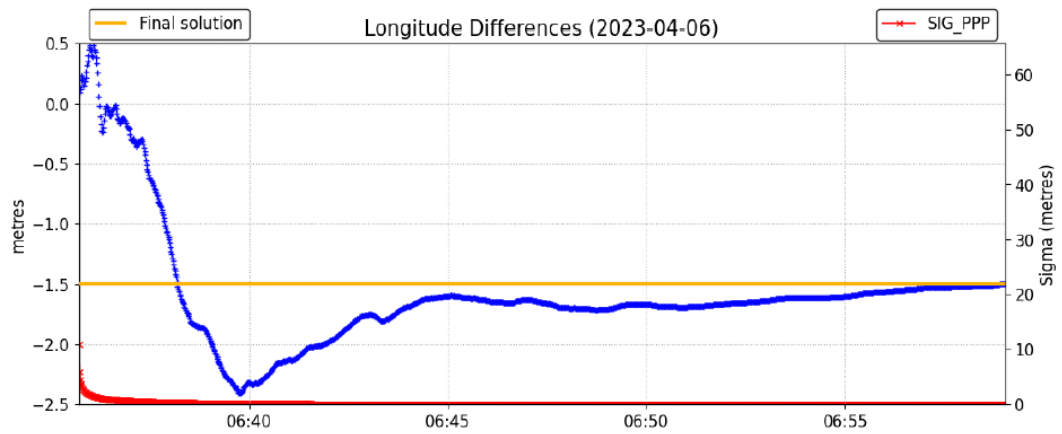


Fig. 4.6: Longitude Variation of Base Station.

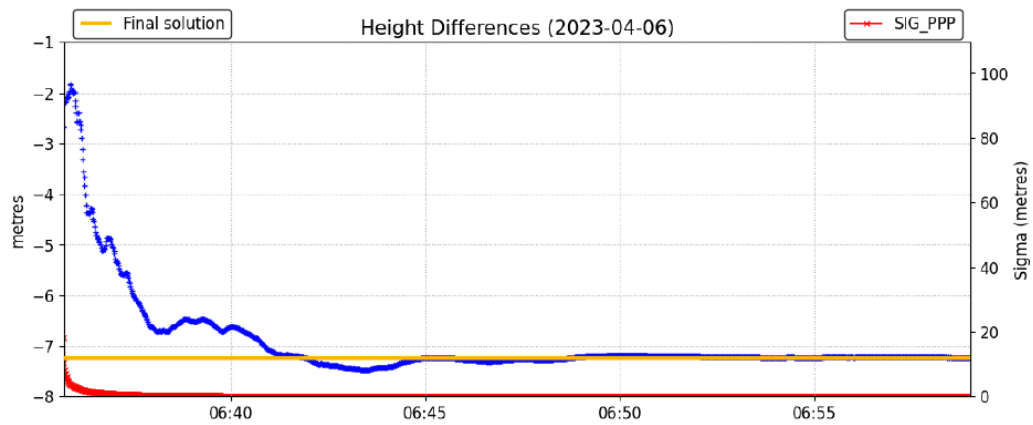


Fig. 4.7: Height Variation of Base Station.

Moreover, the CRS-PPP service station didn't only provide the coordinate correction, but it also helped in the estimation of the tropospheric zenith delays for location of the base station. The result is given as follow:

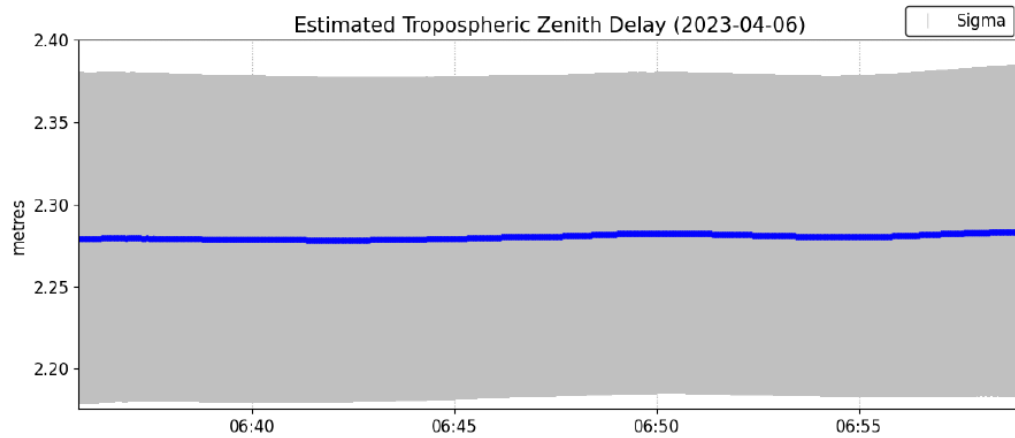


Fig. 4.8: Estimated Tropospheric Zenith Delay

In addition to this, the receive clock offset was calculated too by the same service and its results can be observed in the figure. It can be seen clearly that the clock offset is growing with time.

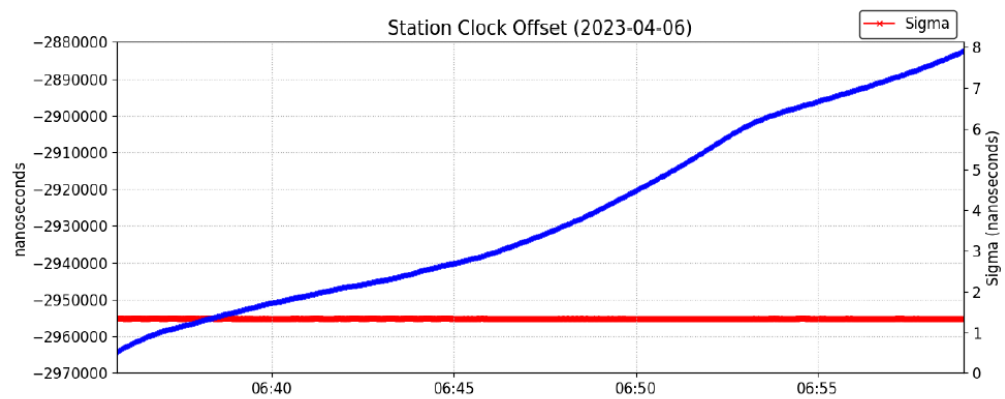


Fig. 4.9: Station Clock Offset

Lastly, the pseudorange and carrier phase residuals were calculated by the same service and the results are given for each satellite tracked.

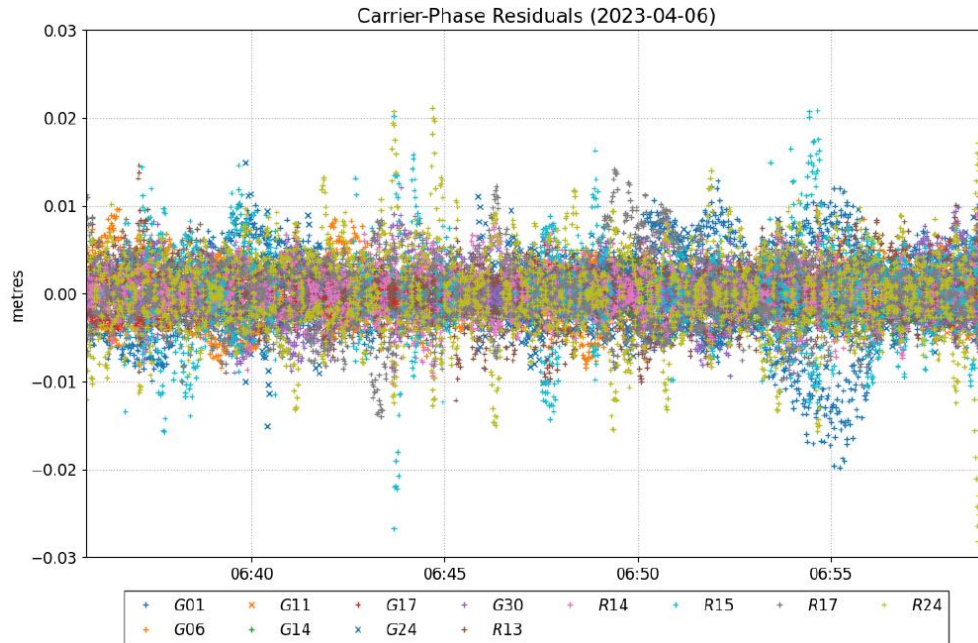


Fig. 4.10: Carrer Phase Residuals.

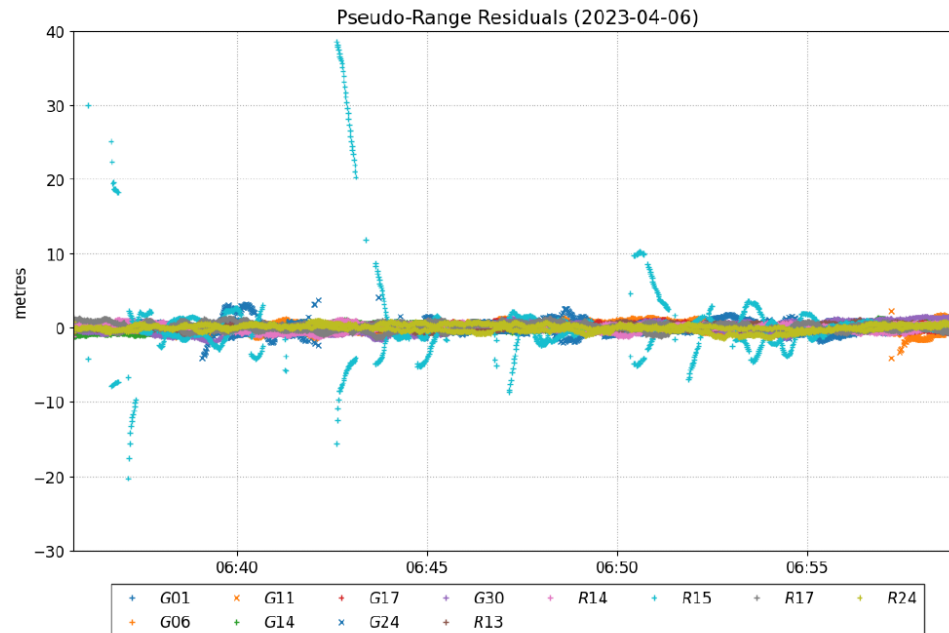


Fig. 4.11: Pseudo Range Residuals

Overall, the duration of observations is 23 minutes and 18 seconds, starting on April 6, 2023, at 06:35:43.00 and ending at 06:59:01.00. The processing time for the service was 5 hours and 27 minutes, and the product type was NRCan Rapid. The observations frequency mode used was Phase and Code Double Static, with an elevation cut-off of 7.5 degrees, and no rejected epochs or fixed ambiguities. The antenna model used was APC to ARP ARP to Marker, with an unknown antenna model. The estimated position for the base station is latitude 33° 31' 14.86700" N, longitude 73° 10' 25.33152" E, and ellipsoidal height 483.563 m, using ITRF20 (2023.3). The sigma (95%) for the estimated position are 0.116 m, 0.327 m, and 0.289 m for latitude, longitude, and ellipsoidal height, respectively. The a priori position is latitude 33° 31' 15.04310" N, longitude 73° 10' 25.38940" E, and ellipsoidal height 490.804 m. The estimated position differs from the a priori position by -5.426 m, -1.494 m, and -7.241 m for latitude, longitude, and ellipsoidal height, respectively.

4.3. Calculation of the Rover Position

Next the position of the rover was calculated as a standalone (single point positioning) using the least square estimates. The results obtained for the standalone position clearly shows that the position is only accurate up to 1 m in horizontal and roughly 2.4 m in vertical.

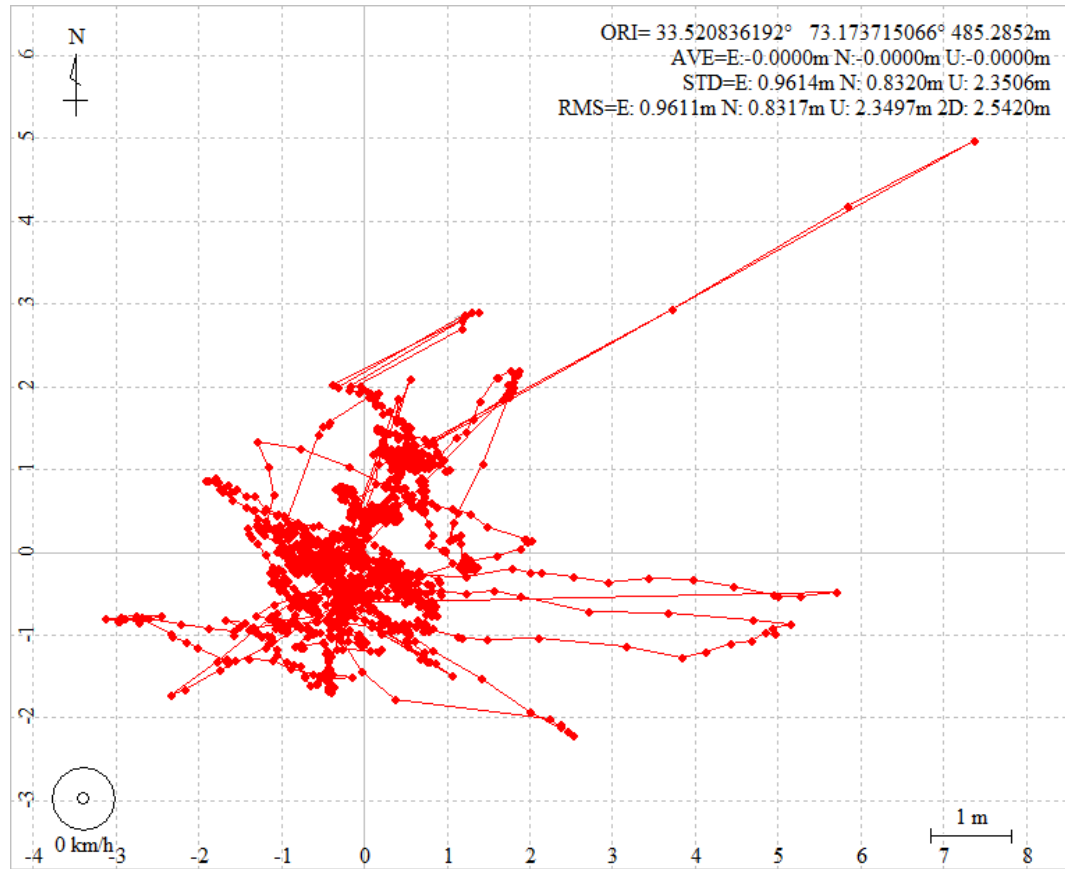


Fig. 4.12: Standalone Position of Rover.

Additionally, it is evident from the figure that there is a clear variation in the coordinates. During the calculation of the stand-alone position, it can be observed that the maximum variation in the E-W coordinate exceeded ± 2 meters at some epochs. On the other hand, in the Up-Down coordinates, the error exceeded ± 5 meters and even reached up to 10 meters at some epochs.

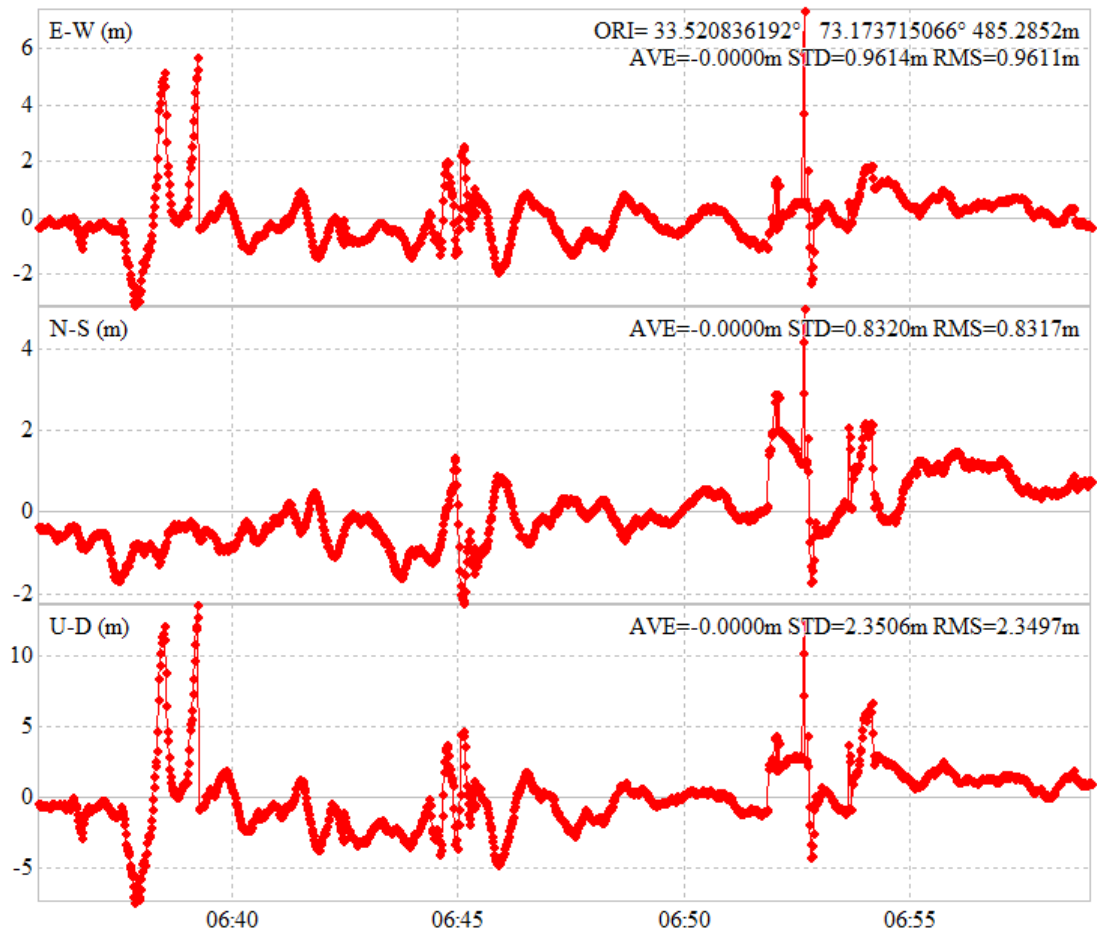


Fig. 4.13: Positioning Variation in East, West and U-D

Such a variation in coordinates may be fine for some applications such a navigation on road, however, it is not suitable for applications that require more accuracy such as guidance and navigation of autonomous vehicles, drones, land surveying and mapping. Under such conditions, there is a need of augmentation service that can help improving the accuracy of the receiver.

4.4. Rover Position Calculation in DGPS Mode

In contrast to the stand alone positioning of the rover station, the following are the results obtained from the DGPS mode or to be more specific, CORS with code based positioning.

Figure shows the position calculated of the rover station with corrections received from the base. The code phase measurements are used for the calculation of the rover receiver position. As seen in the figure results show improvement in position as compared to the standalone positioning.

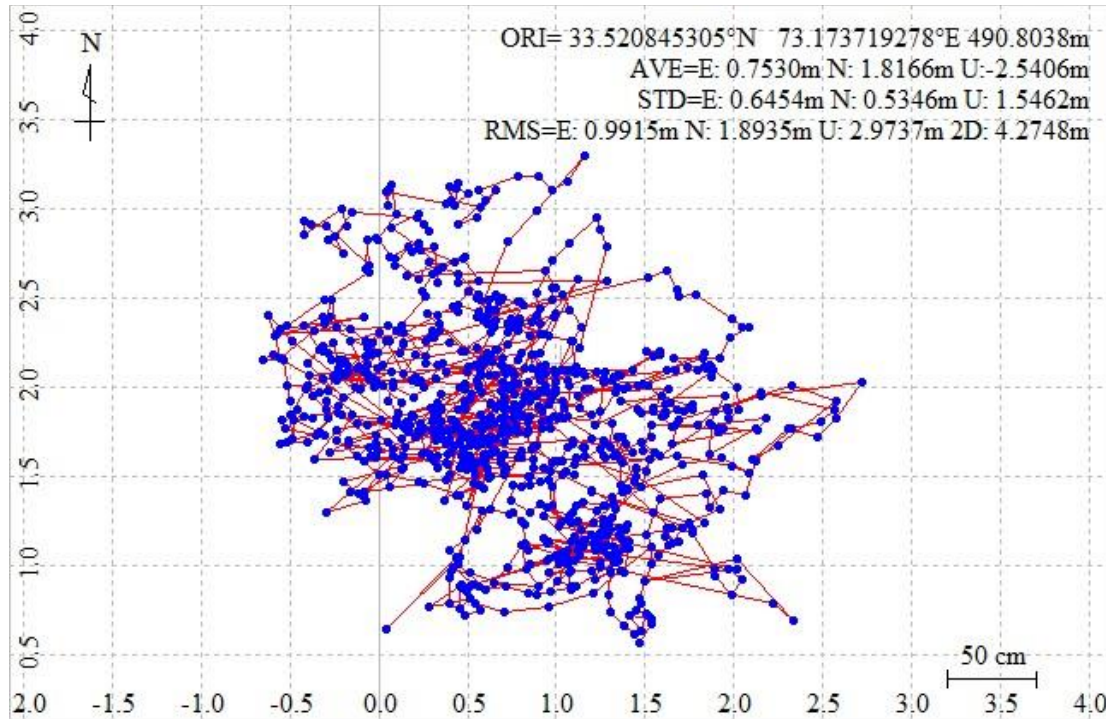


Fig. 4.14: Rover Position in DGPS Mode

As compared to the standalone positioning the standard deviation of the position in east coordinate has improved from 0.9 m to 0.6 m. Similarly, in the North and Up coordinates there is also noticeable improvement of 0.5 m and 1.5 m respectively. Apart from the ground track of the receiver, the variation in coordinates at specific epochs can also be observed from the following figure. In contrast to the standalone mode, it can be noticed that the variation in East-West coordinates only crosses 2 meter at some epochs

while the Up-Down coordinates shows a maximum error of -5 meter at some epochs. This is a significant improvement of accuracy as compared to the single point positioning.

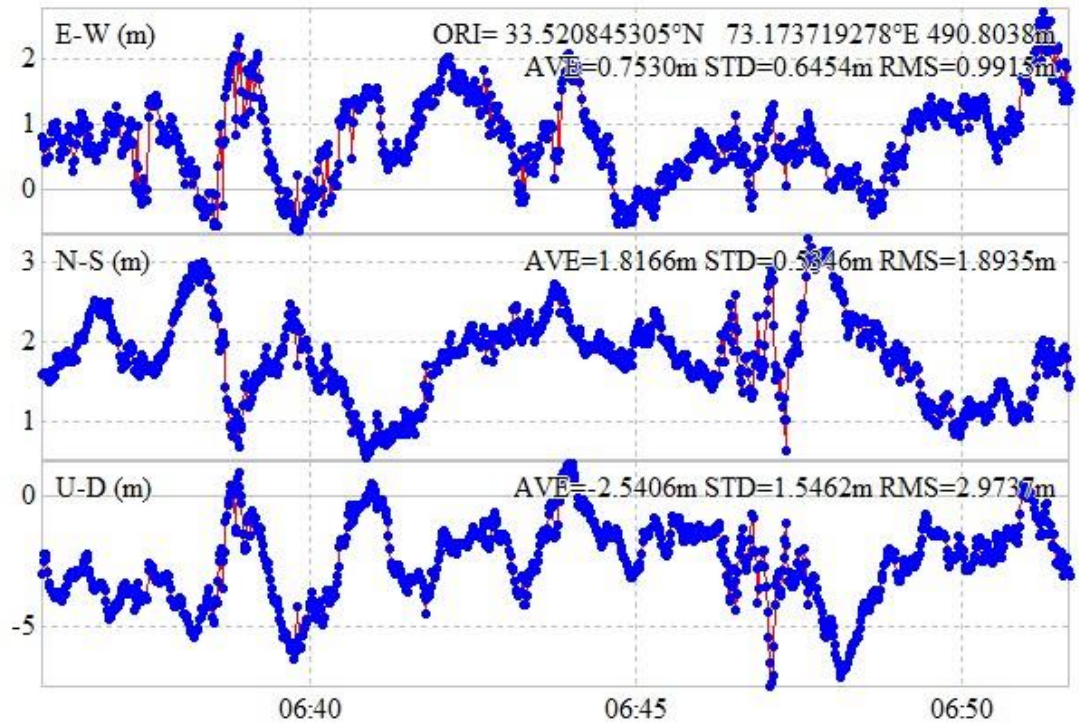


Fig. 4.15: Positioning Variation in East, West and U-D

4.5. Rover Position in RTK Mode – CORS Network

Although the CORS network in code-phase measurements showed an improved accuracy, there is still a gap for improvement that can be filled with the carrier phase based measurements. In contrast to the DGPS – CORS, the RTK – CORS utilizes carrier phase measurement to calculate the position of the receiver. The main difference between the code and carrier phase base measurements is that the wavelength of the carrier is much more precise as compared to the wavelength of the PRN code, therefore, it can achieve

better accuracy with an additional cost of ambiguity resolution. The ambiguity can be resolved with the help of least square ambiguity decorrelation adjustment (LAMDA) method.

Figure 4.16 represents the ground track of the receiver in RTK-CORS mode with fixed solutions only. Float solution means that the ambiguity hasn't been resolved by the receiver yet while the fixed solution means that the receiver has resolved the carrier phase ambiguity. According to the statistical analysis from the given figure the standard deviation in East coordinate has experienced a drastic improvement of up to 0.0021 meter. In addition to this, the variation in East – West, North – South, and Up – Down coordinates can be observed in the Figure which further clarifies the result and shows a promising output of the carrier phase based measurements in terms of positional accuracy.

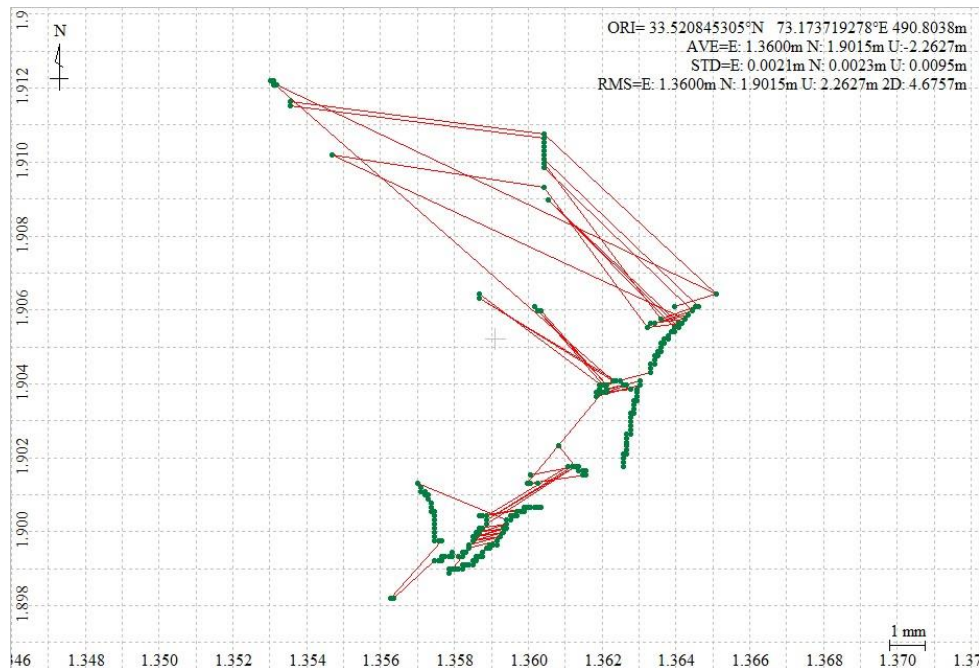


Fig. 4.16: Receiver Ground Track in RTK CORS Mode.

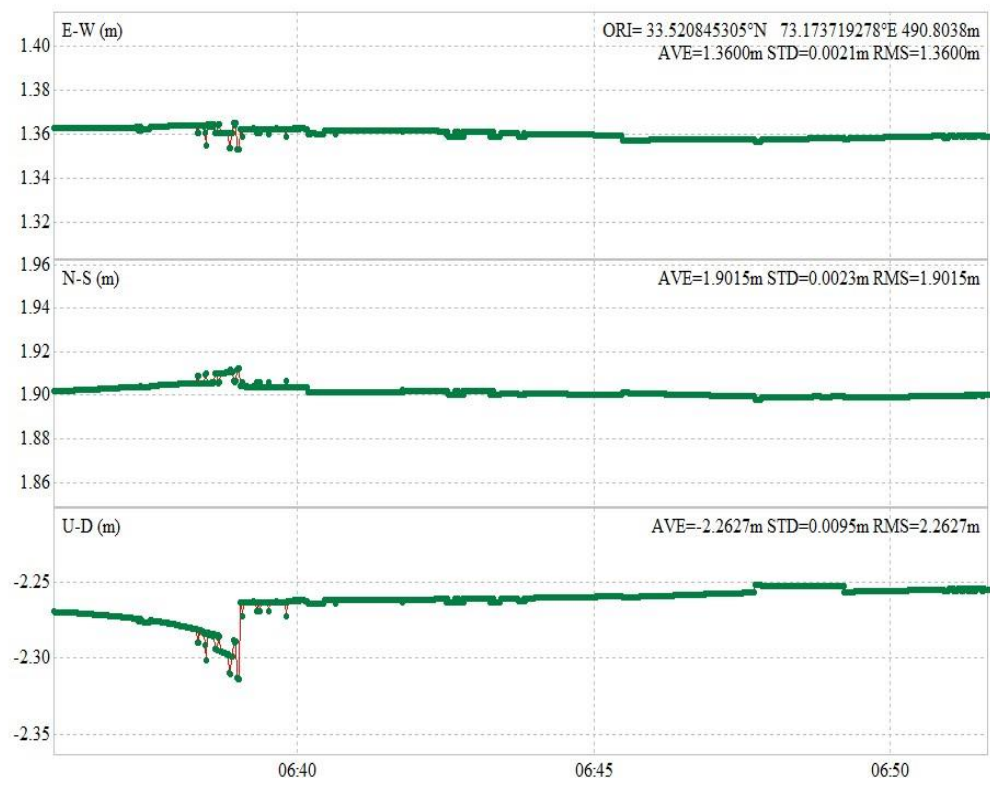


Fig. 4.17: Positioning Variation in East, West and U-D

5. Conclusion & Future Recommendation

The results of this study demonstrate the effectiveness of CORS networks for GNSS-based positioning in large cities and areas where accurate positioning is required for various applications. The use of CORS networks provides a cost-effective and efficient solution for obtaining high-precision positioning results. The obtained results were based on commercial off-the-shelf receivers, which are not typically designed for high-accuracy surveying purposes. During this study multiple experiments were conducted i.e. standalone positioning of the rover station to test its performance in single point positioning. Furthermore, the rover was connected to the base station via a local and a global server to receive its corrections and apply those corrections. The results obtained shows a promising outcome of code based solution but the carrier based output gives even more accurate results with an additional cost of ambiguity resolution.

This study suggests that even higher accuracy results could be achieved with the use of survey-grade or high-accuracy receivers in combination with other accuracy measures. This can be helpful for the navigation of autonomous land and aerial vehicles such as drone based delivery services, self-driving taxis etc. Apart from that, a series of base stations installed throughout the city or a large area will not only be helpful in providing accurate solution for navigation and timing, but it will also be helpful in studying the troposphere of that area using signal from multiple stations. In addition to this, CORS networks can also be quite helpful in studying natural hazards such as earthquakes and Tsunamis. Overall, the use of CORS networks has the potential to revolutionize GNSS-based positioning in urban areas and other regions where high-accuracy positioning is essential.

5.1. Future Recommendation

The current study was limited to the investigation of accuracy improvement with commercial off the shelf receivers and a short-baseline. This study can further be extended to long and very long baselines to test the quality of the corrections provided by the base station to the rover station. Furthermore, the base station can be combined with additional measurement systems such as precise point positioning to further improve the accuracy of the results. This study can be used as a road map to establish a complete regional CORS network which can further be extended to provincial and then the whole country.

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Guideline for Continuously Operating Reference Stations Special

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